

GENERAL SCIENCE GRADE 10: CHEMICAL FAMILIES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.2: Chemical Families
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Alkali metals, alkaline earth metals, halogens, noble gases and transition metals – Physical and chemical properties of selected elements – Uses of the elements and their compounds
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify chemical families in the periodic table, - classify elements of the periodic table into their chemical families, - investigate the physical and chemical properties of selected elements of the periodic table, - outline the uses of elements of the periodic table, - appreciate the uses of the elements of the periodic table in day-to-day life.
Core Competencies	– Self-efficacy: as the learner carries out activities to investigate the reaction of sodium, magnesium and chlorine with air and dilute acids. – Digital literacy: as the learner uses available digital and print media to research on transition elements and their uses.
Core Values	– Love: as the learner takes care of self and others by observing safety when carrying out activities to investigate properties of elements of the periodic table and appropriately using first aid kits in case of injuries.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions on why some elements are used for particular tasks, such as why a jiko sufuria is made of aluminium but wiring of copper, and add them to the Driving Question Board. – Planning and carrying out investigations: learners test physical properties (conductivity, appearance, hardness) of metals and observe reactions of Group 1 and 2 elements with air, water and dilute acids, using teacher demonstrations for the hazardous ones. – Analyzing and interpreting data: learners identify patterns in reactivity down a group and across families from observation tables. – Developing and using models: learners build and revise a family-trends model of the periodic table, linking electron arrangement to reactivity. – Constructing explanations: learners explain why an element is suited to a use, citing evidence from its properties. – Obtaining, evaluating and communicating information: learners research transition elements and noble gas uses from digital and print sources and present in plenary.

Pertinent & Contemporary Issues (PCIs)	– Road safety: as the learner discusses with peers the properties of elements that make them suitable for use in illuminating roads.
Career Connections	– Chemist / laboratory technician: investigating and testing the properties of elements and compounds. – Materials and metallurgical engineer: selecting metals such as copper, iron, zinc and lead for particular uses, for instance in Nairobi's Industrial Area workshops. – Electrician: choosing copper for wiring and understanding properties of metals. – Lighting and road engineer: using noble gases for street and highway lighting. – Water treatment officer: using chlorine to disinfect water, e.g. in Nairobi and Mombasa water works. – Jua kali metalworker: working with aluminium, iron and zinc (galvanised iron sheets) to make sufurias and mabati roofing.
Focus for Lessons	This unit anchors learning in everyday Kenyan objects made from different elements. Learners sort elements into families, investigate their physical and chemical properties safely, and use evidence to explain why elements suit their uses. Each lesson cycles through predict, observe, explain, Driving Question Board and model building, so the family-trends model grows throughout the unit.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life? KICD KEY INQUIRY QUESTIONS: 1. How are elements suited to their use in day-to-day life?
Anchoring Phenomenon	In a Kenyan kitchen and compound, many everyday objects are made of different elements. The sufuria for cooking ugali is aluminium, the electric cable of the jiko or kettle is copper, the mabati roof is iron coated with zinc, drinking water in Mombasa is treated with chlorine, and street lamps along Thika Road glow with sodium light. Yet sodium metal itself reacts violently with water and must be stored under oil, while chlorine is a poisonous gas that makes water safe to drink. Students observe a collection of these objects and a demonstration of a small piece of sodium in water, and wonder why elements with such different, even dangerous, properties each end up as a safe and useful item, and why a given element is chosen for a given job and not another.
Storyline Thread	Lesson 1: Anchoring phenomenon. Students examine Kenyan objects (aluminium sufuria, copper wire, mabati, salt) and watch a sodium demonstration. They predict why different elements are used for different jobs and build the Driving Question Board with initial questions. Lesson 2: Chemical families. Students research and sort the first twenty elements into alkali metals, alkaline earth metals, halogens and noble gases, and note where the transition metals sit. They draft an initial family model. Lesson 3: Physical properties of the families. Students investigate appearance, hardness, density, melting and conductivity of samples and secondary data across families, and record patterns in a table. Lesson 4: Alkali metals. Using teacher demonstrations of sodium (and lithium or potassium if available) with air and water, students observe reactivity, write balanced equations, link reactivity to the single outer electron, and apply safety and first aid rules. Lesson 5: Alkaline earth metals. Students investigate magnesium with air, water and dilute acid, write balanced equations, and compare its reactivity with sodium to revise the model. Lesson 6: Halogens and noble gases. Students study chlorine's reactions (via demonstration, video and simulation) and its use in water treatment in Mombasa and Nairobi, then explain why noble gases are unreactive and suited to road lighting and signs. Lesson 7: Transition metals. Students research and discuss copper, iron, zinc and lead and their uses (wiring, mabati, galvanising, batteries), and match properties to uses with digital and print sources. Lesson 8: Putting it together. Students complete the family model, answer the Driving Question Board, and explain, using evidence, why an element is chosen for a given use, for example by recommending an element for a new Kenyan product. They

	reflect on safety and care for self and others.
Supporting Phenomena	– Table salt (sodium chloride) on our ugali and nyama choma is safe to eat, though it is made from a violently reactive metal and a poisonous gas. – Magnesium ribbon burns with a bright white flame, which is why it is used in flares and fireworks. – Neon and other noble gas signs and lamps glow brightly with colour, yet the noble gases hardly react with anything. – Iron nails rust while zinc-coated mabati roofs last for years, and copper wires conduct current for decades.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes (2 x 40-minute lessons)): The Sufuria, Wire, Mabati, Salt, Chlorine and Sodium-Lamp Puzzle

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle: everyday Kenyan objects made from elements with very different, sometimes dangerous, properties. Learners predict, observe and question why each element suits its job.
Knowledge	By the end of the lesson, the learner should be able to: (a) name the elements in six Kenyan anchor objects (aluminium sufuria, copper wire, mabati of iron coated with zinc, table salt NaCl, Mombasa water treated with chlorine, Thika Road sodium lamp); (b) state that these elements have very different properties and that sodium reacts violently with water while the same element is combined in safe salt; (c) pose questions about why an element suits a particular job.
Skills	Developing and using models (initial ideas); asking questions and defining problems (NGSS SEP 1); planning predictions; observing and recording; constructing explanations from evidence; communicating and collaborating in groups.
Attitudes	Curiosity about how everyday materials work; respect for others' opinions when sharing predictions (Value: Respect); care for self and others by observing safety near reactive elements (Value: Love). PCI: Safety and security, and Road safety (street lighting). Appreciation of the uses of elements of the periodic table in day-to-day life.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	This is the anchoring lesson. It sets up the Driving Question: Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life? Learners start the Driving Question Board (DQB) with the categories Properties, Reactivity, Uses and Safety, and start the class family model (a wall periodic table). Lessons 2 to 8 return to these questions, beginning with Lesson 2 on chemical families and the periodic table layout. No later content (families, outer electron explanations, equations) is taught here.
Safety Notes	Sodium demonstration is by the teacher ONLY (Mr. Otieno model): safety screen, goggles and lab coat, pea-sized piece (about 0.1 g) cut under paraffin oil, dropped into a trough of water; learners stand at least 2 m back behind the screen and wear goggles. Learners never touch or handle sodium. Keep sodium stored under paraffin oil, with no water near the store. Dispose of residues as directed by the teacher and have a first aid kit on hand. Learners handle only the safe objects (sufuria, wire, mabati, salt) with clean hands; no tasting of lab materials. Do not bring chlorine gas to class; use only the water-treatment information or a picture or video.

B. LESSON OVERVIEW

Mama Wanjiru, a food seller in Mombasa, cooks ugali in an aluminium sufuria and asks why her pot is safe while chlorine in her tap water is poisonous as a gas. In this lesson learners examine a collection of Kenyan objects (aluminium sufuria, copper wire from a kettle or jiko cable, mabati of iron coated with zinc, table salt) and look at information on Mombasa water treated with chlorine (about 0.5 mg/L) and Thika Road sodium lamps glowing yellow-orange. They first predict what the sodium in the lamp and salt would do in water, then watch the teacher demonstrate a pea-sized piece of sodium in water: it floats, fizzes, darts about, melts into a ball, and the solution turns pink with phenolphthalein, showing an alkali (NaOH) forms. They are puzzled that this dangerous element is also part of safe salt. In groups they try to explain the puzzle and record questions on sticky notes for the DQB, sorted into Properties, Reactivity, Uses and Safety. They finish by drawing an initial model of why an element suits a job. Learners do not yet explain the puzzle fully; the unit does that.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of four, learners pass round the anchor objects: an aluminium sufuria (from a Gikomba jua kali artisan), copper wire from a kettle or jiko cable, a piece of galvanised mabati, and a bowl of table salt (NaCl). The teacher reads out Mama Wanjiru's question and gives facts: Mombasa water works adds chlorine at about 0.5 mg/L, and Thika Road lamps glow yellow-orange with sodium vapour. Learners write individual predictions in their notebooks: (1) 'What do you think would happen if a small piece of sodium, the element in salt and in the street lamp, were dropped into water?' (2) 'Why do you think each object is made from the element it is made from and not another?' They then share in the group and draw a quick prediction sketch.	Display the objects on a table before the lesson and say: 'Look at these ordinary things from a Kenyan kitchen and compound. Each one is made from a different element. What do you notice?' Hand out the objects and allow 3 minutes. Read Mama Wanjiru's question aloud. Ask: 'Sodium is in this salt, and it is in the Thika Road lamp. Write down what you think a pea-sized piece of pure sodium will do in water.' WAIT TIME 10-15 seconds after the question before anyone speaks. Then cold-call 4 learners from different groups (e.g. Akinyi's group and Kiprop's group) to share predictions without evaluating them: 'Thank you. Tell me what makes you think that.' Record 3-4 predictions on the board under 'Our first ideas'. Say: 'We will not say who is right yet; we will test our ideas.' Circulate and note learners who link properties to jobs, such as the sufuria being light or the wire conducting electricity.	Predict-before-observe with Think-Pair-Share: writing a prediction individually first exposes prior ideas, and sharing in groups lets learners compare them. The prediction is later compared with observations, which creates a need to know and supports the NGSS practice of asking questions and defining problems.	Observe whether each learner writes a prediction with a reason (look for at least one reason per notebook). Check that predictions mention an observable outcome (floats, fizzes, nothing happens, explodes). Listen for misconceptions, such as 'salt is dangerous because it has sodium' or 'all metals sink and are safe', and note them for the DQB. Tally on a checklist how many learners can name the metal in at least 3 of the objects.
Observe Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook)	Learners stand behind the safety screen, wearing goggles, at least 2 m back, while the teacher (Mr. Otieno model) takes sodium from its paraffin oil, cuts a pea-sized piece (about 0.1 g) and shows the shiny cut face before it dulls. The piece is dropped into a trough of water. Learners record what they see in a table: appearance of sodium (soft, silvery, cut with a knife, stored under oil), behaviour in water (floats, fizzes, darts about, melts into a ball), and the colour change when phenolphthalein is	Before the demonstration, say: 'This is a teacher demonstration only. Nobody touches sodium. Goggles on, stay behind the line.' Check that every learner has goggles. Explain why sodium is kept under paraffin oil: 'No water near the store.' Ask: 'What will you look for? Write down three things to observe.' WAIT TIME 10 seconds. Do the demonstration slowly and narrate only what is seen, not why: 'The piece is floating. I can hear fizzing.' Pause after dropping it for learners to	Collecting evidence through a teacher demonstration with a structured observation table (separating observation from inference). This builds the habit of using evidence, and the 'surprise' between a violent element and safe salt generates authentic questions.	Check the observation tables for at least 3 accurate observations of sodium (floats, fizzes, darts about, melts into a ball, pink with phenolphthalein) and that observations are separated from explanations. Note who follows the safety rules (goggles on, 2 m back). Listen for learners expressing surprise or noticing the contrast between sodium and salt or chlorine and safe drinking water.

Source: CK-12 Search ARES for similar readings	added (turns pink, so NaOH, an alkali, has formed). They then examine the anchor data: the sufuria, wire, mabati and salt, a picture of the Mombasa water works and of the Thika Road lamps. They compare the properties of sodium with salt and with the other objects in a 'What I saw / What I know' table.	watch. Add 2 drops of phenolphthalein and ask: 'What colour is it now?' Cold-call 3 learners to describe what they saw using only observations (not explanations). Then say: 'The same element, sodium, is in the salt in Mama Wanjiru's ugali and in the Thika Road lamps. Compare what you saw with the salt in your hand.' Ask: 'What is surprising?' WAIT TIME 15 seconds. Cold-call 2 more learners. Show the chlorine information: 'Chlorine is a poisonous gas, but Mombasa water works adds about 0.5 mg per litre to make our water safe.' Dispose of residues as directed and keep the screen in place until finished.		
Explain Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12 Search ARES for similar readings	In their groups, learners use a 'puzzle map' on a manila sheet to put the six anchor objects (sufuria, wire, mabati, salt, Mombasa water, Thika Road lamp) in a ring around the question 'Why does each element suit its job?'. For each object they write the element (Al, Cu, Fe and Zn, NaCl, Cl, Na), one property they know or think matters (for example light, conducts electricity, does not rust, kills germs, glows yellow) and one thing they wonder. They then try to explain in 2-3 sentences how sodium can be dangerous alone and yet safe as salt, and why chlorine can be poisonous as a gas but safe in water. They compare their first predictions with what they saw and mark them 'supported', 'not supported' or 'not sure'. Groups share their best attempts in a gallery walk and mark ideas they agree or disagree with using different-coloured sticky dots.	Say: 'Your task is to use the evidence you saw to make a first explanation of the puzzle. It is fine to say we are not sure.' Give 10 minutes for group work and circulate with prompts such as: 'What evidence from the demonstration supports that?' and 'Is the property you chose useful for that job? Why?' Do not give the answer (outer electrons and families are for later lessons). If asked 'Why does sodium react so much?' say: 'Great question. Put it on a sticky note for the Driving Question Board; we will investigate it.' Run a gallery walk for 5 minutes. Then ask: 'Which ideas did several groups share? Which ideas are different?' WAIT TIME 10-15 seconds. Cold-call 4 groups to share one idea and one wondering. Record common ideas and disagreements on the board. Highlight that different elements have very different properties and that each is useful in a different	Group puzzle mapping followed by a gallery walk and a claim-evidence-reasoning check: connecting each object to an element, a property and a question helps learners notice patterns across objects. Honest initial explanations, with agree and disagree dots, make gaps in understanding visible and motivate the unit.	Check that puzzle maps correctly name the element in at least 5 of 6 objects (Al, Cu, Fe and Zn, NaCl, Cl, Na) and link at least 3 to a property. Look for explanations that use evidence from the demonstration instead of just opinion. Note the misconceptions to revisit (for example 'salt is just sodium', 'chlorine in water is a gas bubble', or 'dangerous elements are never useful'). Use a thumbs up/side/down check on 'I can explain why sodium in salt is safe' to gauge confidence.

		job, which is the pattern to explain. Praise ideas that use evidence.		
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	Akinyi and Kiprop (DQB keepers for the class) help to set up a wall chart with four columns: Properties, Reactivity, Uses and Safety. Each learner writes at least 2 questions on sticky notes (for example Akinyi: 'Why is sodium so dangerous?', Kiprop: 'Why do street lamps glow yellow?', 'Why is the sufuria made of aluminium and not iron?', 'How can poisonous chlorine make water safe?'). Learners place their notes in the best column and read others' notes. As a class they group similar questions and agree a final Driving Question: 'Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life?' They then choose three questions they most want answered.	Say: 'Good scientists write down what they wonder. Write at least two questions about the puzzle on sticky notes: one question per note, starting with Why or How.' Give 5 minutes. Model one example: 'Why is the mabati coated with zinc?' Circulate and help rephrase yes/no questions into 'why' or 'how' questions. Invite Akinyi and Kiprop to post the first notes and sort the rest into the four columns. Ask: 'Which questions are alike? Which column is missing questions?' WAIT TIME 10-15 seconds. Cold-call 4 learners to read a question aloud and say which column it belongs to. Read out the Driving Question and ask the class to say it in their own words. Explain that the board will be revisited at the end of every lesson, with answers added as evidence is found. Photograph the DQB. Say: 'Today we will not answer them. In the next lesson we look at how the elements are arranged.'	Driving Question Board with categorised student questions: using learners' own questions gives ownership and a visible record of what the class needs to figure out, and sorting them into categories shows the shape of the unit. It supports asking questions and defining problems.	Count questions: each learner has at least 2 on sticky notes. Check that at least half are 'why' or 'how' questions and are placed in sensible columns. Notice whether questions go beyond naming objects, for example linking properties to uses or asking why sodium is reactive. Note learners who have not contributed and invite them with a cold-call or a buddy.
Model Building Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Each learner draws an initial model in their notebook titled 'Why does an element suit its job?'. The model includes: the six anchor objects, the element in each, arrows linking each element to one property and one job, and a box showing sodium and chlorine together becoming salt with a question mark ('what changes?'). They add labels such as 'dangerous alone / safe in salt' and 'unsure' where they cannot explain. They then add to the class family model, a large wall periodic table, by placing sticky labels on	Say: 'Draw your first model. It will be wrong in places, and that is the point. We will revise it in every lesson.' Display an example of a model with arrows from objects to properties. Give 8 minutes. Circulate and ask: 'What does this arrow mean?' and 'What do you want to find out here?' Invite learners to place labels on the class wall periodic table and ask: 'Where do you think each of these elements sits? Does position matter?' (Do not teach families yet.) WAIT TIME 10-15 seconds. Cold-call 3 learners to explain one	Initial model drawing and a class family model on the wall periodic table: representing ideas visually and revising them over the unit lets learners see their own progress. Marking 'unsure' parts turns gaps into investigation goals and supports developing and using models.	Check that models include at least 4 anchor objects with the correct element, at least 3 links from element to property or job, and a visible attempt to show sodium and chlorine in salt. Look for 'unsure' labels as evidence of honest thinking. Use exit tickets to see if learners can state one surprise and one question. Review models after class to plan how to address common ideas in Lessons 2 and 3.

Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12 Search ARES for similar readings	Na, Cl, Al, Cu, Fe and Zn with the job each does (for example Na: street lamp and salt; Al: sufuria). They write a one-line 'What I think now' statement and a 'What I still need to find out' statement.	part of their models. Finish with an exit ticket: 'Write one thing you saw today that surprised you and one question you will check in the next lesson.' Collect models and keep them for revision in later lessons. Link ahead: 'In Lesson 2 we will see how elements are grouped into chemical families in the periodic table.'		
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D. TEACHER REFLECTION

After the lesson, consider: Did every learner see the sodium demonstration clearly and safely from behind the screen? Which predictions were most common, and which misconceptions (for example that salt is the same as sodium) appeared? Did learners ask questions beyond naming objects, such as why sodium is reactive or why chlorine is used in tiny doses? Which DQB columns (Properties, Reactivity, Uses, Safety) had the fewest questions, and how can I encourage more? Were all learners, including quieter ones, heard through cold-calls and wait time? Check that the sodium, paraffin oil and goggles were handled and stored correctly. Plan to open Lesson 2 by returning to the DQB and the wall periodic table.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that everyday Kenyan objects are made from different elements: aluminium sufuria, copper wire, iron mabati coated with zinc, salt (NaCl), chlorine-treated Mombasa water and sodium-lamp light on Thika Road. We also saw sodium float, fizz, dart about and melt in water, and turn the solution pink with phenolphthalein.
What did I learn?	We learned that elements have very different properties, and some, like sodium, are dangerous on their own, yet each is useful and chosen for a job. We now have many questions about why an element suits a job, recorded on the DQB under Properties, Reactivity, Uses and Safety.
How does this explain the phenomenon?	Our first explanation is that each element is used for the job that its properties suit, but we cannot yet explain why sodium is violent alone yet safe in salt, or why chlorine is poisonous as a gas yet safe in tiny amounts in water. We will use the periodic table and later investigations to answer these questions.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Chemical Families: Sorting the First Twenty Elements by Position in the Periodic Table

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners see that elements are sorted into chemical families by their position in the periodic table, and that group number links to similar behaviour. This lets them place the elements behind the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle on a family map and begin predicting which element suits which job.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify chemical families in the periodic table; (b) classify elements of the periodic table into their chemical families. Learners will know that: alkali metals are Group 1 (Li, Na, K), alkaline earth metals are Group 2 (Be, Mg, Ca), halogens are Group 17 (F, Cl), noble gases are Group 18 (He, Ne, Ar), and transition metals sit in the central block from Period 4 (Sc to Zn, including Fe, Cu, Zn). H, B, C, N, O, Al, Si, P and S are not in any of the four families. Elements in the same group have the same number of outer electrons (Group 1 = 1, Group 2 = 2, Group 17 = 7, Group 18 = 8, with He = 2) and so behave similarly.
Skills	NGSS Science and Engineering Practices: Analysing and Interpreting Data (sorting and finding patterns in the first twenty elements table); Developing and Using Models (drafting the initial class family model on the wall periodic table); Constructing Explanations (using position and outer electrons to justify family membership). KICD core competencies: Critical Thinking and Problem Solving (sorting and arranging the first twenty elements according to their similarities); Digital literacy (using available digital and print media to research element categories).
Attitudes	Values: Respect (recognising every group member's input when sorting and debating placements); Unity (taking turns in roles during the card sort). PCI: Life skills (problem solving) as the learner arranges the first twenty elements into groups and periods. Learners appreciate that a position on the periodic table helps predict an element's behaviour in daily Kenyan life.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 1 produced the puzzle (sufuria, wire, mabati, salt, Mombasa chlorinated water, Thika Road sodium lamp, and the sodium-in-water demonstration) and DQB questions about why an element suits a job. Lesson 2 gives learners the first organising tool: the family map of the periodic table. They locate the anchor elements (Al, Cu, Fe, Zn, Na, Cl) on it and draft an initial family model. Lesson 3 will use these families to organise physical property data, and Lessons 4 to 7 will explain family behaviour in detail. This lesson does not explain why families react as they do. It only establishes that position and group number link to similar behaviour.
Safety Notes	This is a paper and wall-chart lesson, so no hazardous chemicals are handled. If the teacher shows sealed samples (Mg ribbon, Al foil, Cu wire, Fe nail, Zn strip), learners wear goggles and handle only those. Sodium, potassium and lithium are NOT handled by learners and are not brought out in this lesson. Remind learners of the Lesson 1 rule: never touch sodium with bare hands, and keep water away from stored sodium (kept under paraffin oil). Keep the aisles clear while learners move to the wall periodic table.

B. LESSON OVERVIEW

Mr. Otieno's Grade 10 class in Kisumu returns to the DQB from Lesson 1, where Akinyi asked why sodium is dangerous and Kiprop asked why street lamps glow yellow. Learners first predict how the elements behind the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle might be grouped. They then research and sort the first twenty elements (symbols, atomic numbers Z and positions) into alkali metals, alkaline earth metals, halogens and noble gases, and note where the transition metals sit. Next they explain why elements in the same group might behave alike, using outer electron counts (Group 1 = 1, Group 2 = 2, Group 17 = 7, Group 18 = 8). They add new questions to the DQB and annotate the large wall periodic table with the anchor elements, producing a first draft of the class family model. Mama Wanjiru's question

(why her aluminium sufuria is safe while chlorine gas is poisonous) is treated as a family question: which families do Al and Cl belong to?

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revisit the Lesson 1 anchor list (aluminium sufuria, copper wire, mabati of iron coated with zinc, table salt, Mombasa chlorinated water, Thika Road sodium lamp). On a half-sheet, individually, they write the symbol and Z of each element: Al 13, Cu 29, Fe 26, Zn 30, Na 11, Cl 17. They then answer: 'If a class of elements had to be sorted into families, which of these six would you put together, and what would you base it on?' They draw their prediction as a quick sketch: a rough blank periodic table with the six elements placed where they think they belong. They share with a partner for 2 minutes. Learners are told that their sketch is a prediction and that it is fine to be wrong.	Display the Lesson 1 anchor objects and the DQB. Say: 'Yesterday we wondered why sodium is dangerous but salt is safe, and why a sufuria and a wire are made of different elements. Today we ask: do elements come in families, like people in a Luo or Kalenjin clan, with shared traits?' Give the six symbols and Z values on the board. Say: 'Silently, for 3 minutes, predict which of these elements might belong together and why.' WAIT TIME: after asking 'What did you base your grouping on?', wait 10-15 seconds before taking answers. Cold-call 4 learners (including Akinyi and Kiprop) using the name cards, and ask each only 'What did you base it on?' Do not correct predictions. Record the bases offered (colour, use, hardness, state, position) on the board under 'Our Ideas for Grouping'.	Predict-Observe-Explain (POE) with individual prediction first. Committing to a written, drawn prediction makes learners' existing ideas visible, so they can later compare them with evidence and notice what they had to revise. Recording the bases for grouping on the board gives the class a shared starting point.	Check the sketches while circulating. Look for: (1) learners who group by use or appearance only (for example, 'sufuria and wire are both kitchen items'); (2) learners who mention position or number; (3) learners who place Na and Cl together because they make salt. Tally on a clipboard how many sketches group by use, appearance or position. Success indicator: at least 4 learners can name a basis for grouping, even if it is incorrect, and every learner has placed all six elements on the sketch.
Observe Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods	In groups of four, learners research and sort the first twenty elements. Each group gets (a) a blank periodic table outline for Periods 1 to 4, (b) 20 element cards (H1, He2, Li3, Be4, B5, C6, N7, O8, F9, Ne10, Na11, Mg12, Al13, Si14, P15, S16, Cl17, Ar18, K19, Ca20) showing symbol, Z and electron arrangement where known (Li 2,1; Na 2,8,1; K 2,8,8,1; Mg 2,8,2; Cl 2,8,7; Ne 2,8; Ar 2,8,8), and (c) a textbook or tablet with a periodic table. Step 1: place	Distribute materials and review the roles: 'Everyone takes a turn. The reader reads aloud, the card handler places cards, the recorder writes, the reporter speaks.' Model the first placement on the board: 'Na is Z = 11, so it comes after Ne at the start of Period 3. Which group is it in?' Circulate to each group at least once. Ask: 'What do you notice about the elements in the same column?' WAIT TIME: wait 10-15 seconds after the question before rephrasing. Probe:	Data sorting and pattern-seeking (structured card sort). Learners physically sort real data into categories, so patterns (family names matching columns, outer electrons matching group numbers) become visible from the data and are not just told. Rotating roles ensures all learners engage in the analysis.	Collect the Family Sorting Table from each group and check against the key. Indicators: alkali metals Li, Na, K placed in Group 1; alkaline earth metals Be, Mg, Ca in Group 2; halogens F, Cl in Group 17; noble gases He, Ne, Ar in Group 18; H, B, C, N, O, Al, Si, P and S marked as not in the four families; transition metals placed in the central block from Period 4. Success indicator: at least 80% of groups sort 16 or more of the 20 elements correctly, and every

and Groups (Flexbook) Source: CK-12 Search ARES for similar readings	the cards in order of Z in the correct rows and columns (groups = vertical columns, periods = horizontal rows). Step 2: use print or digital resources to find the family names for Groups 1, 2, 17 and 18 and mark them with colour pencils (alkali metals, alkaline earth metals, halogens, noble gases). Step 3: sort each of the 20 cards into one of five piles: alkali metals, alkaline earth metals, halogens, noble gases, or 'not in these four families' (H, B, C, N, O, Al, Si, P, S). Step 4: locate the transition metals block between Group 2 and Group 13 beginning at Period 4 (Sc to Zn) and place cards for Fe26, Cu29 and Zn30 in it. Roles: reader, card handler, recorder, reporter, rotated midway. Learners complete a Family Sorting Table with columns: family, group number, elements from the first twenty, number of outer electrons.	'Where did you put H, and why is it not in a family with Li and Na?' Confirm that H is a non-metal that is not placed in any of the four families. Say: 'Aluminium is Z = 13, in Period 3, and it is a metal, but it is not in any of our four families. Mark it as other.' Stop groups at the 25-minute mark. Cold-call 5 groups to read out one family each from their Family Sorting Table: 'Group 3, read us the alkali metals.' Ask the class to show thumbs up or down to check. Then ask a group to show where the transition metals sit. Clarify that Pb (Z = 82) is beyond the first twenty and is not sorted today. Remind learners that the anchor object elements will be located on the table in the next phase.		group correctly states that transition metals start at Period 4. Note groups that misplace Al or H, and reteach at their table.
Explain Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Learners work in the same groups, then share with the whole class. Part A (10 minutes): They use their Family Sorting Table to answer: (1) What do the elements Li, Na and K have in common in their electron arrangements (2,1; 2,8,1; 2,8,8,1)? (2) What is the number of outer electrons in Mg and Ca (Group 2)? Cl (Group 17)? Ne and Ar (Group 18)? (3) What pattern links group number to outer electrons (Group 1 = 1, Group 2 = 2, Group 17 = 7, Group 18 = 8, He = 2 as a full first shell)? Part B (10 minutes): Each group writes a claim-evidence-reasoning statement: 'Claim: Elements in the same group are likely to behave similarly. Evidence: ... Reasoning: ...' Part C (8 minutes):	Ask: 'Look at your Family Sorting Table. What is the same for every element in one column?' WAIT TIME: wait 10-15 seconds, then allow pair talk for 1 minute. Cold-call 3 pairs and write their ideas on the board: same number of outer electrons, same group number, similar position. Then say: 'Group 1 elements each have one outer electron. Group 2 has two, Group 17 has seven, Group 18 has eight, except helium, which has two because its first shell is full.' Ask: 'What might the number of outer electrons have to do with how an element behaves?' WAIT TIME: wait 15 seconds. Accept ideas as hypotheses and say: 'We will test this idea in the next lessons. Today we only note the link	Claim-Evidence-Reasoning (CER) with discourse. Writing a claim, tying it to table evidence and giving reasoning forces learners to connect position, group number and outer electrons, rather than memorise family names. Pair talk before cold-calling lets every learner rehearse an answer first.	Read the written CER statements. Indicators: a correct claim that same-group elements behave similarly; evidence citing at least two specific elements and their outer electron counts (for example Li 2,1 and Na 2,8,1 each have 1 outer electron); reasoning that links group number to similar behaviour. Check that Na, Cl, Al, Fe, Cu and Zn are all located correctly. Success indicator: at least 75% of learners write a CER statement with a correct claim and at least one correct piece of evidence. Misconception to watch: learners who say all metals are in one family, or who say hydrogen belongs to Group 1 as an alkali metal.

 Search ARES for similar readings	Using the anchor data, they locate the six elements of the puzzle on their table and name their family or block: Na is an alkali metal (Group 1); Cl is a halogen (Group 17); Al is a metal not in the four families (Group 13); Fe, Cu and Zn are transition metals. They answer Mama Wanjiru's question in one sentence: 'Her sufuria is aluminium and chlorine is a halogen, so they belong to different parts of the periodic table.' Learners note that they cannot yet say why those positions give those behaviours; that is for later lessons.	between group number and outer electrons.' Do not explain reactivity or why Na is dangerous; say: 'Hold onto Akinyi's question about sodium. Lesson 4 will answer it.' Give the CER frame on the board. Check 3 claim statements by cold-calling and asking each speaker: 'What is your evidence from the table?' For Part C, ask: 'Where does copper sit? Which block?' and 'Is aluminium in one of our four families? How do you know?' Praise careful use of the words 'group', 'period' and 'chemical family' and correct 'row' and 'column' confusion: group is vertical, period is horizontal.		
Driving Question Board (DQB) Creation  VIDEO: Electron configurations for the third and fourth periods <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Writing the Periodic Table (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the class DQB, divided into the four columns Properties, Reactivity, Uses and Safety. Each learner writes one new question on a sticky note based on today's findings, for example: 'Why do elements in the same family behave alike?' (Reactivity), 'Are all alkali metals as dangerous as sodium?' (Safety), 'Why is aluminium not in a family like sodium?' (Properties), 'Is chlorine like other halogens?' (Reactivity), 'Why are the noble gases used in signs?' (Uses), 'Why is copper in a block with iron?' (Properties). They place the notes in a column and cluster similar questions. Akinyi and Kiprop, the DQB keepers, review old Lesson 1 questions. Learners put a tick on any Lesson 1 question that today's family map partly helps with (for example Kiprop's question on the yellow lamp now has a clue: sodium is an alkali metal). They do not mark any question as answered yet.	Say: 'Take 2 minutes to write one new question you have now that you did not have before.' WAIT TIME: wait 10-15 seconds after the instruction before anyone moves. Cold-call 4 learners to read their question aloud and say which column it belongs in: 'Which column does your question go in, and why?' Ask Akinyi and Kiprop to cluster similar notes and read one cluster title. Ask: 'Which old questions from Lesson 1 can we now say something about?' Cold-call 2 learners. Point out that most answers are still missing and that the DQB will guide the next lessons: physical properties (Lesson 3), alkali metals (Lesson 4), and so on. Do not answer any questions today. Say: 'Good questions are more valuable than quick answers.'	Driving Question Board with question clustering. Learners generate and sort questions from new evidence, so they see how today's learning leads to new puzzles and owns the direction of the unit. Linking old questions to the new family map shows progress without prematurely answering.	Collect the sticky notes and check quality. Indicators: each question is specific to families or to the anchor elements (not 'what are elements?'), is placed in a sensible column, and shows a link between position and behaviour or use. Success indicator: at least 80% of learners produce a question that mentions a family, group, or one of the anchor elements. Tally how many questions fall in each column and note which families learners are most curious about, to guide Lessons 3 and 4.

Model Building Phase  VIDEO: Electron configurations for the second period <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mendeleev's Periodic Table (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners build the first draft of the class family model on the large wall periodic table. In turns, each group sticks coloured labels on the wall chart for the first twenty elements and the transition block: orange for alkali metals, blue for alkaline earth metals, green for halogens, purple for noble gases, grey for transition metals. They then annotate the six anchor elements with small picture cards: Al (sufuria), Cu (wire), Fe and Zn (mabati), Na (salt and Thika Road lamp), Cl (Mombasa water), and write the family or block name under each. Beside the chart, each learner copies a personal model into their notebook with a key. Below it they complete a short statement: 'In my family model, elements in the same group have the same number of _____ outer electrons, so I predict they will _____. ' They add a 'What we still don't know' box listing at least two open questions (for example why sodium reacts violently with water, why noble gases are used in lamps, which properties make Al suit the sufuria).	Say: 'This wall table is our family model. It is a first draft, and we will revise it every lesson.' Hand each group two element labels at a time so all groups get a turn, with WAIT TIME of 10-15 seconds before each group places its labels so peers can check. Cold-call 3 learners to challenge a placement: 'Is Al correctly placed? How do you know?' Remind: 'Aluminium and hydrogen are not in the four families. Mark them differently.' As Akinyi and Kiprop keep the model, ask them to read the labels aloud. Prompt: 'Use the model to predict: will potassium behave more like sodium or like chlorine? Say why in one sentence.' Cold-call 3 learners. Accept predictions as ideas to test, for example 'like sodium, same group, one outer electron'. Do not confirm reactivity details. Explain homework: find one more object at home or in the market (for example a jua kali item from Gikomba, a battery, or a nail) and name one element in it.	Cumulative model building. Learners draft a visible model that links position, family and outer electrons to the anchor objects, and then use it to make a prediction. Because the model is annotated each lesson, they can see it grow and can identify gaps (the 'don't know' box) that drive later investigation.	Check the notebook models and the wall chart. Indicators: all 20 elements and the transition block placed in the correct colour-coded family; the six anchor elements correctly located; H and Al marked as not in the four families; a prediction that uses group number (for example K will behave like Na). Success indicator: at least 80% of learners have a correct model and one reasoned prediction, and each lists at least two open questions. Use the 'What we still don't know' boxes to plan Lesson 3 on comparing physical properties.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners sort the first twenty elements using position and group number rather than guessing from the names? Which groups misplaced H or Al, and why? (2) Could learners state the link between group number and outer electrons without being led, or did they need prompting? (3) Did I resist explaining why Na is dangerous and keep that for Lesson 4? (4) Did the cold-calling and 10-15 second WAIT TIME give quieter learners, including those who sat at the back, a chance to speak? (5) Were the new DQB questions linked to families and the anchor objects? (6) What misconceptions emerged (for example that all metals form one family, or that hydrogen is an alkali metal)? Plan to address them in Lesson 3 with the physical property data. Note learners who need extra support with reading the periodic table and arrange pairing with a confident partner.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We sorted the first twenty elements by position on the periodic table. Li, Na and K are in Group 1 (alkali metals); Be, Mg and Ca
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	are in Group 2 (alkaline earth metals); F and Cl are in Group 17 (halogens); He, Ne and Ar are in Group 18 (noble gases). H, B, C, N, O, Al, Si, P and S are not in any of these four families. The transition metals, including Fe, Cu and Zn, sit in the central block starting from Period 4.
What did I learn?	Elements are grouped into chemical families by their position in the periodic table. Groups are vertical columns and periods are horizontal rows. Elements in the same group have the same number of outer electrons (Group 1 = 1, Group 2 = 2, Group 17 = 7, Group 18 = 8, with He = 2), which links group number to similar behaviour.
How does this explain the phenomenon?	Sodium (Na, Thika Road lamp and salt) is an alkali metal, chlorine (Cl, Mombasa water) is a halogen, aluminium (sufuria) is a metal not in the four families, and copper (wire), iron and zinc (mabati) are transition metals. So the elements in our puzzle come from different parts of the periodic table, and a family's position can help us predict behaviour. We still need evidence on their properties and reactivity to explain why each element suits its job.

LESSON 3 (80 minutes (2 x 40-minute lessons)): Physical Properties of the Families: Finding Patterns in Density, Melting Point, Hardness and Conductivity

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners compare the physical properties of elements from different chemical families, using samples and a secondary data table, to find patterns across and within families. This gives them evidence for why an element suits a job in the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle.
Knowledge	By the end of the lesson, the learner should be able to: (a) compare hardness, density, melting point and conductivity for selected elements from the alkali metals, alkaline earth metals, transition metals, halogens and noble gases; (b) identify patterns, e.g. Na, K and Li are soft, low-density, low-melting metals that float on water (1.00 g/cm ³), while Mg, Ca and the transition metals are denser and melt at higher temperatures; (c) state that metals in the data table conduct electricity while Cl ₂ , Ne and Ar do not; (d) connect a property to a job, e.g. copper's excellent conductivity for wire and aluminium's low density for the sufuria.
Skills	Science and Engineering Practices: (1) Planning and carrying out investigations: testing hardness, density and conductivity of Mg, Al, Cu, Fe and Zn samples with goggles on. (2) Analysing and interpreting data: organising the secondary data table by family and finding patterns. (3) Using mathematics: calculating density as mass divided by volume and comparing values with water at 1.00 g/cm ³ . (4) Constructing explanations from evidence. Core competencies: Critical thinking and problem solving; Communication and collaboration.
Attitudes	Learners appreciate that evidence from careful measurement lets us predict which element is right for which job. Values: Respect (listening to group members' interpretations of the data) and Unity (taking assigned roles in the group practical). Learners also show care and responsibility for themselves and others when handling samples.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 2 organised the first 20 elements into chemical families by position in the periodic table. This lesson uses those family groupings to organise physical data and search for patterns. It prepares Lesson 4 (why alkali metals are soft, light and very reactive), Lesson 5 (Mg is denser and higher melting than Na) and Lesson 7 (transition metals as strong, dense, high-melting conductors). Learners start to see why sodium is stored under oil and cut with a knife, why aluminium suits a sufuria and why copper suits wire. Reasons based on outer electrons are left for Lessons 4 to 6.
Safety Notes	Goggles on for all practical work. Learners handle only Mg ribbon and Al, Cu, Fe and Zn samples. Na, K and Li are never handled by learners. Mr. Otieno shows them only as sealed, labelled samples under paraffin oil, or uses a photo or video. No water near stored sodium. Chlorine is treated as secondary data only; no chlorine gas is prepared. Use low-voltage dry cells for the conductivity circuit and keep it away from water. Handle the cut edges of metal strips carefully. Dispose of samples as the teacher directs, and keep the first aid kit on the bench.

B. LESSON OVERVIEW

Mr. Otieno opens with Mama Wanjiru's question from the Mombasa market: why is her aluminium sufuria safe on the fire while the sodium in the demo was dangerous? Learners first predict how the families differ in density, melting point, hardness and conductivity. In the Observe phase, groups test real samples of Mg, Al, Cu, Fe and Zn for appearance, hardness (scratch test), density (mass and water displacement) and conductivity (simple circuit). They then study a secondary data table covering Na, K, Li, Mg, Ca, Fe, Cu, Zn, Al, Pb, Cl₂, Ne and Ar. In the Explain phase, learners sort the data by family and identify patterns within and across families. Akinyi and Kiprop add new questions to the Driving Question Board (DQB) under 'Properties', and learners annotate the class family model with property labels. Learners finish with an

initial model linking family position to physical properties and to jobs such as sufuria, wire and mabati.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the anchor objects (aluminium sufuria, copper wire, mabati roof of iron coated with zinc) and the sodium demo from Lesson 1. Working individually for 3 minutes, then in pairs, they complete a prediction table with four rows: density, melting point, hardness and conductivity. For each row they predict whether sodium (Group 1), magnesium (Group 2), copper (transition), aluminium and chlorine would be high or low, and write one sentence of reasoning, e.g. 'Sodium was cut under oil with a knife, so I predict it is soft.' They also answer: 'Sodium floated in the trough. What does floating tell us about its density compared with water at 1.00 g/cm ³ ?'	Mr. Otieno holds up the sufuria, a piece of copper wire and a strip of mabati and says: 'Mama Wanjiru cooks ugali in this aluminium sufuria every day. Last lesson we grouped elements into families. Today I want your best guess: do elements in the same family look and behave alike when we compare them physically?' He gives 3 minutes of silent individual writing, then 2 minutes for pairs. He then says: 'Think about the sodium we saw floating. What does floating tell you?' WAIT TIME of 10-15 seconds before taking answers. He cold-calls 4 learners, using no hands up, to share one prediction and reasoning, and records each prediction on the board without correcting it. He asks: 'Who disagrees with that, and why?' Pairs must give a reason, not only an answer. Predictions are photographed or copied for return in the Explain phase.	Eliciting prior ideas through a prediction table. Putting ideas in writing before any evidence is shown commits learners to a testable claim, so they can later compare evidence with their own thinking. Learners link everyday observations to scientific ideas such as density and floating, which strengthens connections to the anchoring phenomenon.	Observable indicators: (1) at least 80% of learners have a complete prediction table with reasoning, not just answers; (2) learners link floating sodium to density lower than 1.00 g/cm ³ ; (3) Mr. Otieno notes common ideas such as 'all metals are hard and heavy' or 'all metals have the same melting point' for later testing; (4) Mr. Otieno tallies how many predict sodium is softer than copper. Learners who write only one-word answers get a prompt: 'What did you see in the demo that makes you say that?'
Observe Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Part A (about 25 minutes): Hands-on testing. Groups of four, with goggles on, take roles: materials manager, tester, recorder and reporter. They test Mg ribbon and Al, Cu, Fe and Zn samples. (1) Appearance: colour and shine. (2) Hardness: scratch each sample with a nail or the edge of a coin and rank them from softest to hardest. (3) Density: measure mass on a balance, find volume by	Mr. Otieno begins with a safety check: 'Goggles on before you touch anything. Show me.' He reminds learners that only the five metal samples are handled and that sodium, potassium and lithium are teacher-only. He circulates and asks each group: 'How will you make sure your scratch test is fair?' (same pressure, same tool) and 'How do you know your density reading is reliable?' (repeat	Data collection and representation through a structured results table. Learners gather first-hand and secondary evidence on the same properties, so they can compare their own measurements with accepted values and judge reliability. Using several kinds of evidence (hands-on testing, data tables and a sodium sample or video) builds a more robust base for finding patterns than any one	Observable indicators: (1) learners use a fair scratch test and record hardness as a ranking, not just 'hard' or 'soft'; (2) density calculations are correct (mass divided by volume, units g/cm ³) in at least 4 of 5 groups; (3) learners correctly say which metals float on water using the 1.00 g/cm ³ benchmark: Na, K and Li float, Mg, Ca, Al and the transition metals sink; (4) learners record

Physical Properties (Flexbook) Source: CK-12 Search ARES for similar readings	water displacement in a measuring cylinder, calculate mass divided by volume in g/cm³ and compare with water at 1.00 g/cm³; they predict, then check, whether each sample sinks. (4) Conductivity: connect each sample in a circuit with a dry cell and bulb or buzzer and record whether it lights. They record results in a table. Part B (about 15 minutes): Secondary data. Learners receive the physical data table: Na: 0.97 g/cm³, 98 C, soft, conducts. K: 0.86 g/cm³, 63 C, soft, conducts. Li: 0.53 g/cm³, 181 C, soft, conducts. Mg: 1.74 g/cm³, 650 C, harder, conducts. Ca: 1.55 g/cm³, 842 C, harder, conducts. Fe: 7.87 g/cm³, 1538 C, hard, conducts. Cu: 8.96 g/cm³, 1085 C, hard, excellent conductor. Zn: 7.14 g/cm³, 420 C, conducts. Al: 2.70 g/cm³, 660 C, conducts. Pb: 11.34 g/cm³, 327 C, soft and heavy, poor-to-moderate conductor. Cl₂: greenish-yellow gas, melts -101 C, boils -34 C, non-conductor. Ne: colourless gas, boils -246 C, non-conductor. Ar: colourless gas, boils -186 C, non-conductor. Mr. Otieno also shows sealed sodium under paraffin oil, or a photo or video of it being cut with a knife, to link back to the demo.	the measurement). He asks one reporter per group, 5 groups in total, to read their density values aloud and posts them on the board. He then says: 'Compare your measurements with the data table. Are they close? Why might they be slightly different?' (impurities, measuring errors). WAIT TIME of 10-15 seconds before taking answers. For Part B he says: 'Sodium sank neither in oil nor in water; it floated. Look at the density column. Which other metals would float on water? Turn to your partner.' He cold-calls 3 pairs and asks: 'What evidence in the table supports that?' He adds: 'Notice there is no melting point row for the gases, only a boiling point. Why do you think that is?' He does not explain the pattern yet; he lets learners notice first.	source alone. Reading the table for sodium, magnesium and copper links each family directly to the objects from Lesson 1.	conductivity accurately; (5) the teacher checks that every learner in each group handled or recorded something. Quick check question: 'Which has the higher melting point, K or Fe?' (Fe, 1538 C versus 63 C). Learners who struggle with the density calculation get a worked example with the Al sample.
Explain Phase  VIDEO: Atomic radius trends on periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners sort the data table by chemical family using the groupings from Lesson 2: alkali metals (Li, Na, K), alkaline earth metals (Mg, Ca), transition metals (Fe, Cu, Zn), halogen (Cl₂) and noble gases (Ne, Ar); Al and Pb are marked 'other metals' for now. In groups of four they identify at least two patterns: (a) within a family, e.g. Li, Na and K are all soft, low-density metals that float, with melting points of 181, 98 and	Mr. Otieno says: 'Using the family groupings from last lesson, sort this table. Where do the elements in the same family look alike, and where do they differ?' He gives 8 minutes of group work and circulates, asking: 'What exactly in the table makes you say that?' Pushes: 'Is it true for every element in that family?' (check against the data, e.g. Pb is soft yet very dense). WAIT TIME of 10-15 seconds after each question. He	Pattern finding through data sorting and Claim-Evidence-Reasoning. Sorting the data by family shows learners that position in the periodic table predicts physical properties, which advances the driving question. CER forces every claim to be tied to specific numbers, so learners practise arguing from evidence rather than opinion. Revisiting predictions builds the habit of revising ideas based on data.	Observable indicators: (1) learners state at least two accurate patterns using numbers from the table; (2) CER statements include a claim, at least two data points and a reason that links the property to the job; (3) learners notice the exceptions (Pb soft yet dense; Al not in the four families); (4) learners mark predictions as supported or changed with a reason. Exit question on a slip: 'Name one family and two physical

 EXERCISE: Physical Properties of Matter Practice Source: CK-12  Search ARES for similar readings	63 C, so melting point falls from Li to K; (b) across families, e.g. the alkali metals are soft, with low density and low melting points, while the transition metals are hard, dense and high melting; (c) all the listed metals conduct electricity and the three gases (Cl₂, Ne, Ar) do not; (d) the gases have very low melting and boiling points. They then complete a claim-evidence-reasoning (CER) statement, e.g. 'Claim: copper is better than sodium for electric wire. Evidence: copper is hard, ... conducts excellently; sodium is soft with a melting point of 98 C. Reasoning: ...' They return to their Predict-phase table and mark each prediction as supported, changed or not supported. Finally, they match properties to jobs: Al to sufuria (low density at 2.70 g/cm³, conducts heat), Cu to kettle and jiko cable (excellent conductor), Fe and Zn to mabati.	cold-calls 5 groups to state one pattern each and writes them on a class chart without repeating any. He asks: 'Does the pattern hold in the other direction? If a metal is soft, is it always low density?' (No: lead is soft and heavy at 11.34 g/cm³.) He asks: 'Is melting point a good way to tell elements apart across families?' He then says: 'Return to your predictions. Which ones did the evidence change? Be ready to say why.' He cold-calls 3 learners. He introduces Mama Wanjiru's question: 'She asks why her aluminium sufuria is safe. What physical properties make aluminium suitable?' He accepts answers about low density, conducting heat and not melting at cooking temperature (660 C). He clarifies that explaining WHY the alkali metals are so soft and reactive comes in Lesson 4, so he does not give the electron explanation today.	Noting exceptions such as lead guards against overgeneralising.	properties that most of its elements share.' Common error to watch for: stating that all metals are hard (Na, K, Li and Pb are soft). Teacher re-teaches with the table if more than a quarter of learners hold this idea.
Driving Question Board (DQB) Creation  VIDEO: 04 Criteria For Purity, Introduction To Melting Points <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12	Akinyi and Kiprop lead the DQB. Learners write new questions on sticky notes about what they have found and post them under the headings 'Properties', 'Reactivity', 'Uses' and 'Safety'. Examples: 'Why are Li, Na and K all soft and low density?' 'Why does melting point fall from Li to K?' 'Why do the gases not conduct electricity?' 'Why is copper a better conductor than lead?' 'Why did the Thika Road lamp use sodium if it is so reactive?' They also review earlier questions and mark any that the data table partly answers, e.g. 'Why does sodium float?' (density 0.97 g/cm³, below water at 1.00 g/cm³). Each learner places at least one note, and the class sorts the questions into those	Mr. Otieno says: 'Take two minutes of silent thinking. What do you still wonder about the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle after today's data?' WAIT TIME of 10-15 seconds before learners start writing. He asks Akinyi and Kiprop to read out the new notes and place them on the board with the class. He asks: 'Which question can we already partly answer? Which needs more evidence?' He cold-calls 3 learners to explain why they placed a note under a given heading. He steers learners toward questions about reasons ('why does it follow this pattern?') and shows that these will be taken up in Lessons 4 to 6. He says: 'Let us keep our promise to the DQB.'	Question tracking through the Driving Question Board. Moving from 'what do we see?' to 'why does it happen?' gives learners ownership of the next steps and links today's data to the driving question. Sorting questions into answered and unanswered shows how evidence builds across lessons and how new questions arise as old ones are answered.	Observable indicators: (1) every learner posts at least one note; (2) at least half the notes move beyond description to 'why' questions; (3) learners place notes under the correct headings and can justify the placement; (4) learners identify at least one earlier question that today's evidence has partly answered. Teacher note: record the number of 'why' questions as a gauge of readiness for Lesson 4 on alkali metals and outer electrons.

Search ARES for similar readings	answerable with today's data and those that need new evidence from Lessons 4 to 7.	We will return to these notes in the next lesson.'		
Model Building Phase  VIDEO: Basic Physical Properties - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos  HTML: Physical Properties (Flexbook) Source: CK-12 Search ARES for similar readings	Learners add to the class Family Model, the large wall periodic table, and to their own notebook models. For each family they annotate with property labels: alkali metals (Li, Na, K): soft, low density, float on water, low melting points, conduct; alkaline earth metals (Mg, Ca): harder and denser than alkali metals, higher melting points, conduct; transition metals (Fe, Cu, Zn): hard, dense, high melting points, good conductors (Cu excellent); halogen (Cl₂): greenish-yellow gas, non-conductor; noble gases (Ne, Ar): colourless gases with very low boiling points, non-conductors. Each learner draws a simple model in the notebook showing a family on the table with arrows linking its physical properties to one Kenyan job, e.g. Cu to the kettle cable, Al to the sufuria, Fe and Zn to the mabati roof. Learners add one unanswered 'why' for each family in a column labelled 'Still to explain'.	Mr. Otieno says: 'Our wall periodic table is our Family Model. Come and add one property label for a family, and say your evidence.' He calls 5 learners, one per family, and asks the class: 'Does the evidence support that label? Show me thumbs up or sideways.' WAIT TIME of 10-15 seconds before votes. He asks: 'What does this model let us predict? If I give you an unknown soft metal with density 0.9 g/cm³, which family might it belong to?' He cold-calls 3 learners. He reminds them that the model is not finished: 'We still cannot explain why the families differ, or why sodium reacts so violently. Which lessons will help?' He prompts them to draw a link between family position and use, and sets the learning intention for Lesson 4: why alkali metals are so reactive.	Cumulative model building. Learners annotate a shared model and also draw their own, so they can see how the evidence from this lesson fits into the bigger picture. Using the model to predict an unknown element tests whether the idea of family patterns really works. Listing what the model cannot yet explain keeps learners aware that models are revised as new evidence comes in.	Observable indicators: (1) each learner's model correctly labels at least three families with two physical properties each; (2) the model includes at least one link from properties to a Kenyan job; (3) learners predict the likely family of an unknown soft, low-density metal as an alkali metal; (4) the 'Still to explain' column has at least one realistic question. Teacher checks notebooks while circulating. If more than a quarter of learners mislabel a family property, the teacher revisits the data table at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, Mr. Otieno should consider: (1) Did learners use numbers from the table, rather than general statements, to describe patterns? (2) Did the hands-on measurements of density and conductivity match the data table closely enough for learners to trust the evidence? If not, what caused the differences? (3) Which misconceptions appeared, e.g. 'all metals are hard' or 'a heavier-looking metal always has a higher melting point', and were they resolved by the evidence? (4) Did every learner take part in the group practical, and were roles shared fairly between girls and boys? (5) Were the safety rules followed throughout, with goggles on and sodium, potassium and lithium handled only by the teacher? (6) Which DQB 'why' questions are most common, and how should Lesson 4 start from them? (7) How well did learners connect each property to a Kenyan job such as the sufuria, kettle cable or mabati roof?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tested Mg, Al, Cu, Fe and Zn for appearance, hardness, density and conductivity, and studied a data table for Na, K, Li, Mg, Ca, Fe, Cu, Zn, Al, Pb, Cl ₂ , Ne and Ar. Na, K and Li are soft with densities below water (0.97, 0.86, 0.53 g/cm ³) and low melting points (98, 63, 181 °C). Mg, Ca, Fe, Cu and Zn are denser and melt at higher temperatures. The metals conduct electricity; Cl ₂ , Ne and Ar do not.
What did I learn?	Elements in the same chemical family share similar physical properties, and families differ from each other. Alkali metals are soft, low in density and low melting. Transition metals are hard, dense, high melting and good conductors. Position in the periodic table helps us predict physical properties, though there are exceptions such as soft, dense lead.
How does this explain the phenomenon?	Properties help explain why an element suits a job: aluminium is light and conducts heat for the sufuria, copper conducts electricity excellently for wire, and iron is strong for the mabati base. Sodium is soft, melts at only 98 °C and floats, which helps us understand why it behaves differently from the metals we use for pots and wires. We still need to explain why the families differ and why sodium reacts so strongly with water, which we will investigate in Lesson 4.

LESSON 4 (80 minutes (2 x 40-minute lessons)): Alkali Metals: Why One Outer Electron Makes Sodium Dangerous Alone but Safe in Salt

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain why alkali metals are very reactive, and why reactivity increases down Group 1 ($\text{Li} < \text{Na} < \text{K}$), by linking observed reactions with water to the single outer electron that is easily lost. They write balanced equations and apply safety and first aid rules for reactive metals.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify the alkali metals as Group 1 elements with one outer electron (Li 2,1; Na 2,8,1; K 2,8,8,1); (b) describe the reactions of sodium, lithium and potassium with water and air; (c) write balanced equations $2\text{Li} + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2$, $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$, $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$ and $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$; (d) explain that reactivity rises down the group because the outer electron is lost more easily ($\text{Li} < \text{Na} < \text{K}$); (e) state safety rules and first aid for reactive metals.
Skills	Developing and using models (electron arrangement diagrams); planning and carrying out investigations (observing a teacher demonstration); analysing and interpreting data (ranking reactivity from evidence); constructing explanations; using mathematics (balancing equations); communicating (classroom discourse).
Attitudes	Love: care for self and others by observing safety and using first aid appropriately (KICD value). Self-efficacy: confidence in carrying out and interpreting investigations on reaction of metals with air and water (KICD core competency). Respect for peers' ideas during discussion. Appreciation of why safe handling matters in Kenyan labs, workshops and homes.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 1 showed that sodium reacts violently with water yet salt is safe. Lesson 2 placed sodium in Group 1. Lesson 3 showed Na, K and Li are soft, low-density and low-melting. Lesson 4 gives the first reactivity evidence and the first explanation (one easily lost outer electron), answering Akinyi's DQB question 'Why is sodium dangerous?'. This idea is reused in Lesson 5 (Mg, two outer electrons, less reactive), Lesson 6 ($\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$) and Lesson 8.
Safety Notes	TEACHER DEMONSTRATION ONLY for sodium, lithium and potassium; learners never handle them. Mr. Otieno wears goggles and a lab coat, works behind a safety screen, and uses only a pea-sized piece (about 0.1 g) of sodium, with smaller pieces of Li and K, cut under paraffin oil. Learners stay at least 2 metres back, wearing goggles. No water near stored sodium; keep metals in paraffin oil. Use a large trough, no lid, and dispose of residues as the teacher directs (do not tip leftover metal in the sink). Keep a bucket of dry sand, a first aid kit and water source ready. First aid: for skin contact, brush off any metal and flood with running water for at least 10 minutes; for eye splashes, rinse with clean water for 10 minutes and get medical help; report every injury. Phenolphthalein and alkaline solutions are irritants, so wear goggles when handling them.

B. LESSON OVERVIEW

Mr. Otieno's Kisumu class returns to the sodium puzzle from Lesson 1. Learners first predict how lithium, sodium and potassium will compare in water and why. They then watch a teacher demonstration behind a safety screen and record observations in a table (floating, fizzing, speed, melting, flame, colour with phenolphthalein). Next they rank reactivity ($\text{Li} < \text{Na} < \text{K}$), draw the electron arrangements (Li 2,1; Na 2,8,1; K 2,8,8,1) and explain with the idea that an atom with one outer electron loses it easily, and more easily as the outer electron is further from the nucleus. They write and balance equations and rehearse safety and first aid rules. Akinyi and Kiprop update the Driving Question Board (Properties, Reactivity, Uses, Safety) and the class family model, adding the Group 1 reactivity arrow. The lesson ends with a link back to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle: sodium is dangerous alone because it loses its outer electron so easily, so we must understand

it before judging where it is safe.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Visually understanding balancing chemical equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balancing Chemical Equations Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the sodium demo from Lesson 1 and Group 1 from Lesson 2. Individually, in 3 minutes, each writes a prediction on a sticky note or in their notebook: 'If a pea-sized piece of lithium, sodium and potassium each go into water, what will happen, and which will react fastest? Why?' They draw a quick sketch of the three troughs. In pairs they compare predictions, then add one reason, using the soft, low-density, low-melting properties from Lesson 3 (e.g. 'they float because density is below 1.00 g/cm ³ ').	Display the three symbols and Z (Li 3, Na 11, K 19) and the Group 1 column on the wall periodic table. Say: 'Remember Mama Wanjiru's question: her aluminium sufuria is safe, but sodium reacts violently with water. Today we compare three cousins in Group 1. Before I show anything, what do you think will happen? Write your prediction first.' Give 3 minutes of silent writing, then WAIT TIME 10-15 seconds after asking: 'Which one do you think will be the most dramatic, and what is your reason?' Cold-call 4 learners from different parts of the room, no hands up, and record each prediction on the board in two columns: 'Prediction' and 'Reason'. Do not say who is right. Say: 'Keep your note. We will test it in a few minutes.' Circulate and note misconceptions such as 'heavier means faster' or 'the metal burns in the water because water is hot'.	Eliciting prior ideas through commit-and-compare predictions. Writing a prediction with a reason before evidence makes learners' mental models visible, creates a personal stake in the demonstration, and lets them later revise their own thinking rather than just receive an answer.	Observable indicators: (1) at least 80% of learners write a prediction AND a reason; (2) listen for use of Lesson 3 ideas (soft, low density, floats) and Lesson 2 ideas (same group, so similar). Tally how many predict K fastest versus Li fastest. Record misconceptions on a clipboard checklist to address in the Explain phase. Learners who give only a prediction without a reason get a prompt: 'What makes you think so?'
Observe Phase  VIDEO: Visually understanding balancing chemical equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners sit at least 2 metres back, wearing goggles, behind the safety screen. They watch Mr. Otieno (a) cut a fresh piece of sodium under paraffin oil and note the shiny surface that quickly dulls in air, and (b) drop a pea-sized piece of sodium, then lithium, then potassium (if available) into separate troughs of water with phenolphthalein. In groups of 4 they complete an observation table: metal, floats/sinks, fizzing	Before starting, rehearse safety: 'Goggles on. Everyone stays behind the line. No one touches the metals. Why do we use only a pea-sized piece?' Cold-call 2 learners for answers. Cut sodium under oil with a knife, show the shiny surface for 5 seconds, and ask: 'What do you notice about the cut surface and what do you think happens to it in air?' WAIT 10-15 seconds, cold-call 2 learners. Before each drop ask: 'Based on	Structured observation with data tables (Science and Engineering Practice: Planning and carrying out investigations). Separating observation from explanation keeps learners focused on evidence, and the common table lets groups compare the same variables across three metals so a pattern (Li < Na < K) can emerge from data rather than from the teacher's statement.	Observable indicators: tables are completed for all three metals with specific descriptive words (not just 'it reacted'); learners record the pink colour and link it to an alkali; at least one entry per group notes that the metals float (link to density below 1.00 g/cm ³). Check that groups rank the three metals by speed and violence of reaction. Teacher checks goggles on and distance kept for every learner throughout. Ask one group

 HTML: Hydrogen and Alkali Metals (Flexbook) Source: CK-12  Search ARES for similar readings	(slow/fast/very fast), melts into a ball (yes/no), flame colour, solution colour with phenolphthalein. Roles: observer 1 (movement), observer 2 (sound and fizz), recorder, timer who estimates seconds until the reaction slows. Learners see: Li slow fizz; Na fast fizz, melts and darts about, solution turns pink; K very fast, lilac flame. If K or Li are not available, learners use the teacher's described reference data card for these two and clearly mark it as 'secondary data' in the table. After the demo, groups add a note on the metal stored under paraffin oil and why (no air, no water).	your prediction, what will you watch for?' Drop lithium first, then sodium, then potassium, pausing 30 seconds between each to let learners record. After each, ask the recorder group to read their observation aloud. Say: 'Describe only what you SEE and HEAR, not why yet.' Add a few drops of phenolphthalein to each trough afterwards and ask: 'What does the pink colour tell us about the solution?' WAIT 10 seconds; cold-call 2 learners and confirm that an alkali has formed. Point to the bubbles: 'A gas is made. We will find out what it is in the Explain phase.' If any metal remains, place it in the dry sand or dispose as the teacher directs, and state this aloud as modelling safe practice. Do not hurry or add larger pieces.		member to read their table; correct vague statements with 'Can you describe what you saw in more detail?'
Explain Phase  VIDEO: The periodic table - classification of elements <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Hydrogen and Alkali Metals Practice Source: CK-12  Search ARES for similar readings	Learners use their data to rank reactivity: Li < Na < K. They draw electron arrangements in their notebooks and on the board: Li 2,1; Na 2,8,1; K 2,8,8,1, and circle the single outer electron in each. In groups they build a claim-evidence-reasoning (CER) statement: 'Alkali metals are very reactive because ...' and 'Potassium is more reactive than sodium because ...' using the idea that each atom loses its outer electron easily, and the outer electron in K is further from the nucleus (more shells) so is lost more easily. They then write word equations and balance symbol equations: $2\text{Li} + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2$; $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$; $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$; and $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (the dulling of cut sodium in air). They identify the gas as hydrogen, H₂, and explain why the solution turned pink	Return to the predictions board: 'Look at your sticky note. Whose prediction matched? What do you want to change?' Cold-call 3 learners. Ask: 'Which metal was the most reactive? What evidence from your table shows that?' WAIT 10-15 seconds, cold-call 3 groups, and write the order Li < Na < K on the board. Draw the three atoms on the board and ask: 'What is the same about all three? What is different?' WAIT 10 seconds. Guide: all have one outer electron; K has more shells. Ask: 'If an atom has just one electron on the outside, what might it want to do with it?' WAIT 15 seconds; cold-call 2 learners. Introduce only the idea that the metal atom LOSES its outer electron (do not teach ionic bonding in detail, ion formulas, or electronegativity). Say: 'The further the outer electron is from the nucleus, the easier it is	Claim-Evidence-Reasoning (CER) with model-based explanation (Science and Engineering Practices: Constructing explanations; Developing and using models; Using mathematics). Learners must link their own observed evidence to an atomic-level model of outer electrons, which makes the invisible cause of reactivity concrete and turns balanced equations into a record of what happens rather than a memorised routine. Revisiting predictions supports conceptual change.	Observable indicators: (1) correct ranking Li < Na < K supported by at least two pieces of evidence from the table; (2) electron arrangements written correctly (Li 2,1; Na 2,8,1; K 2,8,8,1) with the outer electron circled; (3) CER statements include the single outer electron AND the idea that it is lost easily; (4) at least 75% of pairs balance all three water equations correctly; (5) learners name H₂ and NaOH/LiOH/KOH correctly; (6) safety card correctly lists at least four rules and the first aid steps. Exit check: on a slip, 'Why is potassium more reactive than lithium?' Learners who say only 'because it is bigger' are given the prompt 'What does that mean for the outer electron?'.

	(alkali: hydroxide formed). They also complete a safety and first aid card: rules (goggles, small quantities, never touch with bare hands, store under paraffin oil, no water near stored sodium, dispose as directed) and first aid steps for skin, eye and fire situations, and link these to the earlier sodium risk in Lesson 1. Finally they compare their original predictions with the evidence and write one sentence: 'My prediction was ... The evidence showed ... so I now think ...'.	to lose, so reactivity goes up the way we saw: Li, then Na, then K.' Model balancing $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ on the board by counting atoms on each side aloud, then let learners balance the Li and K equations in pairs while you circulate. Cold-call 3 pairs to write their balanced equations on the board; ask the class to check each by counting atoms. Ask: 'Is this equation balanced? How do you know?' Address misconceptions such as 'sodium burns the water' or 'bigger atom, slower reaction'. Run a safety card activity: 'Mr. Otieno says never touch sodium with bare hands. Use the first aid steps to explain why and what you would do if a splash reached your arm.' Cold-call 4 learners across the room, then confirm the steps. Close by linking back: 'Sodium is dangerous alone because its one outer electron is lost so readily. That is also why, once combined, as in salt on Mama Wanjiru's ugali, it is a very different substance. We will explore that with chlorine in Lesson 6.'		
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Akinyi and Kiprop lead the DQB update. In their groups, learners write new questions and answers on sticky notes, sorting them into the DQB categories Properties, Reactivity, Uses and Safety. They move Akinyi's question 'Why is sodium dangerous?' to 'partly answered' with a note: 'one outer electron, lost very easily'. They add new questions such as 'Why does magnesium not react with cold water like sodium?', 'Why is sodium safe in salt?', 'Why do street lamps along Thika Road glow yellow?' and 'What do we do	Say: 'Akinyi and Kiprop, please read the questions on the board that we can now partly answer.' Give groups 4 minutes to write at least 2 new questions each. Ask: 'Which questions did today's evidence help us answer, and which new ones did it raise?' WAIT 10-15 seconds, then cold-call 4 groups to place one note and explain the category chosen. Prompt: 'Which new question links to the periodic table position?' Point out that questions on Mg and halogens will be tackled in Lessons 5 and 6 without	Driving Question Board as a public record of growing understanding (question-driven learning). Sorting questions into categories shows learners how today's evidence answers some questions, narrows others and generates new ones, so learning feels purposeful and connected to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle.	Observable indicators: every learner contributes at least one question or answer note; notes are placed in suitable categories; at least two new questions show reasoning from today's evidence (for example comparing Na with Mg, or asking why sodium in salt is safe). Teacher checks the Safety category contains at least one first aid question. Note learners whose questions are only factual recall and encourage 'why' or 'how' questions.

Balancing Chemical Equations Source: PhET Interactive Simulations (English) Search ARES for similar readings	if sodium touches skin?'. Each learner contributes at least one note, and the class decides which questions the next lesson (Lesson 5) can address.	answering them yet. Ensure Safety notes receive attention: 'Who can add a question or a rule about first aid?' Photograph or copy the DQB for the offline record.		
Model Building Phase  VIDEO: Ionization energy: group trend <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Hydrogen and Alkali Metals (Flexbook) Source: CK-12 Search ARES for similar readings	Learners update the class family model, the large wall periodic table. They add to Group 1: the label 'Alkali metals: 1 outer electron', the electron arrangements Li 2,1; Na 2,8,1; K 2,8,8,1, and an arrow down the group labelled 'reactivity increases: $\text{Li} < \text{Na} < \text{K}$ '. They add a small note 'very reactive, stored under paraffin oil, teacher demo only' and the key equation $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. In their notebooks, each learner revises their own model of 'Why is sodium dangerous alone?' with a labelled diagram: sodium atom 2,8,1, the outer electron leaving, water, and hydrogen gas bubbles. They finish with a one-sentence link: 'Because sodium atoms lose their single outer electron very easily, ... so sodium is stored under paraffin oil and ... but I still need to find out why salt (NaCl) is safe.' They complete the summary table prompt (observed, learned, explained).	Invite two learners to add the labels and arrow to the wall periodic table while the class checks: 'Does this match our evidence? Is the arrow in the right direction?' WAIT 10 seconds. Ask: 'Predict: where would you place rubidium below potassium? What reactivity would you expect and why?' Cold-call 2 learners (extension, no equipment). Ask: 'What is still missing from our model?' WAIT 10-15 seconds; cold-call 3 learners; guide toward: why Mg is different (Lesson 5), why sodium combined with chlorine is safe (Lesson 6), why sodium vapour glows yellow. Say: 'Keep your model; we will keep revising it.' Collect notebook diagrams at the door and check for the outer electron and the hydrogen gas. For homework, ask learners to find two Kenyan places where reactive metals must be stored or handled carefully (for example school labs or jua kali workshops in Gikomba) and write one safety rule for each.	Cumulative model revision (Science and Engineering Practice: Developing and using models). Adding the Group 1 reactivity trend and electron arrangements to a shared model lets learners see that position in the periodic table predicts behaviour, which is the central idea for choosing the right element for the right job, and gives a concrete anchor for later lessons.	Observable indicators: wall model annotations are accurate (1 outer electron, arrow direction, correct electron arrangements); individual notebook diagrams show the sodium atom with the outer electron leaving, the hydrogen gas and the correct balanced equation; learners can state a limit of the model (it does not yet explain why salt is safe or why Mg is less reactive). Score each diagram: electron shown, trend shown, equation balanced, safety named. Provide extra support for learners missing two or more.

D. TEACHER REFLECTION

After the lesson, reflect on these questions: (1) Did every learner stay behind the safety line and wear goggles, and were safety rules followed without reminders? (2) How many learners moved from predicting by 'size or weight' to explaining by 'outer electron lost more easily', and what evidence from CER statements shows this? (3) Which learners struggled to balance $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ and what helped (counting atoms aloud, drawing atoms)? (4) Did the observation table help learners separate what they saw from why it happened? (5) If potassium or lithium was unavailable, did the secondary data card give learners enough evidence to rank $\text{Li} < \text{Na} < \text{K}$? (6) Did the DQB capture new 'why' questions, and are the Lesson 5 and Lesson 6 questions ready? Note any misconceptions (for example that sodium 'burns' water,

or that a bigger atom is less reactive) to revisit at the start of Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Lithium fizzes slowly in water, sodium fizzes fast, melts into a ball and darts about, and potassium reacts very fast with a lilac flame. All three float and the solution turns pink with phenolphthalein. Freshly cut sodium is shiny but quickly dulls in air. (Record what you saw and heard in your own words.)
What did I learn?	Alkali metals (Group 1) have one outer electron (Li 2,1; Na 2,8,1; K 2,8,8,1). Reactivity increases down the group: $\text{Li} < \text{Na} < \text{K}$. They react with water to form an alkali (hydroxide) and hydrogen gas, and with oxygen in air. They must be stored under paraffin oil, handled only by the teacher, and any contact with skin or eyes needs immediate flushing with running water.
How does this explain the phenomenon?	Alkali metals are very reactive because each atom has a single outer electron that is lost very easily. Down the group the outer electron is further from the nucleus, so it is lost more easily and reactivity increases. This is why sodium reacts violently with water ($2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$) and is dangerous alone, which links to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle. We still need to explain why sodium combined as salt is safe.

LESSON 5 (80 minutes (2 x 40-minute lessons)): Alkaline Earth Metals: Is Magnesium as Dangerous as Sodium?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners compare how magnesium reacts with air, water and dilute acid, relate its lower reactivity to its two outer electrons, write balanced equations for Mg, and use the evidence to revise the class family model and explain why Mg and Na are suited to different jobs in the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle.
Knowledge	By the end of the lesson, the learner should be able to: (a) investigate the physical and chemical properties of selected elements of the periodic table (magnesium with air, water and dilute acid); (b) state that alkaline earth metals (Group 2: Be, Mg, Ca) have two outer electrons (Mg 2,8,2); (c) explain that magnesium is less reactive than sodium (order $K > Na > Li > Ca > Mg$) even though both lose outer electrons; (d) write balanced equations $2Mg + O_2 \rightarrow 2MgO$ and $Mg + 2HCl \rightarrow MgCl_2 + H_2$; (e) appreciate the uses of the elements of the periodic table in day-to-day life.
Skills	Developing and using models (revising the class family model); planning and carrying out investigations (air, water and acid tests with Mg ribbon); analysing and interpreting data (ranking reactivity from observations); constructing explanations; using mathematics (balancing atoms); core competency: Self-efficacy as the learner carries out activities to investigate the reaction of magnesium with air and dilute acids.
Attitudes	Love: learners take care of self and others by observing safety when investigating the elements and by knowing where the first aid kit is. Respect: learners listen to and build on group members' ideas. Responsibility: learners handle materials and waste as directed. PCI: Safety and security.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 4 showed that alkali metals are very reactive because each has one easily lost outer electron. Here learners test a Group 2 metal and find it far less violent than sodium. This gives the first evidence that reactivity differs between families and that outer electrons alone do not decide it. It also begins to answer Akinyi's DQB question on why sodium is dangerous, and prepares Lesson 6 (halogens and noble gases).
Safety Notes	Goggles on for all learners at all times. Mg ribbon pieces about 2 cm only. Teacher demonstrates the burning of Mg in air with a tongs, behind a safety screen, and learners must NOT look directly at the bright white flame (view through a gap in fingers or at an angle, or use a piece of dark glass). Dilute HCl about 0.5 mol/dm ³ only: wash splashes with plenty of running water; first aid kit and water available. Hydrogen gas is flammable: no naked flame near the acid test tubes except the teacher's supervised pop test. Learners do not handle sodium, potassium or lithium; any reference to Na is from Lesson 1 and 4 evidence. Dispose of residues as directed by the teacher.

B. LESSON OVERVIEW

Mr. Otieno in Kisumu reminds the class of the sodium demo and the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle. Akinyi asks: 'If sodium is so dangerous, are all metals like that?' Learners predict how magnesium ribbon will behave compared with sodium. They then gather evidence in groups: Mg ribbon in cold water, Mg ribbon in dilute HCl, and a teacher demonstration of Mg burning in air. They compare with the Lesson 4 sodium evidence and the Lesson 3 data (Mg density 1.74 g/cm³, melting 650 C versus Na 0.97 g/cm³, 98 C). In the Explain phase they use the electron arrangements Na 2,8,1 and Mg 2,8,2 to build explanations, write and balance $2Mg + O_2 \rightarrow 2MgO$ and $Mg + 2HCl \rightarrow MgCl_2 + H_2$, and rank $K > Na > Li > Ca > Mg$. They add new questions to the DQB and revise the class family model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Alkaline Earth Metals Practice Source: CK-12  Search ARES for similar readings	Learners are shown a 5 cm piece of shiny Mg ribbon, beside a photo of the sodium demo from Lesson 1. Individually (3 minutes) on a prediction sheet they write: (1) what will happen if Mg ribbon goes in cold water, (2) what will happen in dilute HCl, (3) what will happen if it is heated in air, and (4) whether Mg will be more or less reactive than Na, with a reason using what they know about outer electrons from Lesson 4. They draw a quick sketch of what they expect to see in the water test. Then they pair-share and mark each prediction with a confidence level (sure / not sure).	Hold up the ribbon and say: 'Remember Mr. Otieno's sodium fizzing across the trough on Thika Road day. Magnesium sits next to sodium in the periodic table. Will it do the same? Write your prediction first, no talking.' Give WAIT TIME of 10-15 seconds after each question before anyone speaks. Walk the room and note 4 different predictions. Cold-call 3 learners (not volunteers) to share one prediction and its reason. Say: 'Thank you. I am not telling you who is right. We will let the evidence decide.' Write 2 competing predictions on the board. Point out that the magnesium used in school labs is also in some ordinary things, such as flash bulbs in older cameras and in some antacid tablets, so learners see it is not always dangerous.	Prediction with confidence rating (commitment before evidence): writing a prediction and a reason makes learners' current model visible and gives them a stake in the outcome. When evidence later contradicts a prediction, the conflict drives revision of the model, not just memorising a result.	Look for: (a) at least 3 of 4 prediction items completed with a reason; (b) whether the learner links reactivity to outer electrons or just guesses ('because it is a metal'); (c) at least 5 of 8 groups include a comparison with Na. Record the common misconception that 'all metals next to sodium must fizz in water' to revisit in the Explain phase.
Observe Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Alkaline Earth Metals (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of 4 (roles: reader, handler, recorder, reporter) wearing goggles do Station A and Station B for 12 minutes each, then watch the demo. Station A (cold water): clean a 2 cm piece of Mg ribbon with sandpaper, drop it into a test tube of cold water, observe for 3 minutes, then add 2 drops of phenolphthalein; record bubbles, colour and whether the ribbon moves or melts. Station B (dilute acid): place a 2 cm clean Mg ribbon in a test tube with 5 cm ³ of dilute HCl; record bubbling speed, temperature change by feeling the outside of the tube, and what happens to the ribbon. The teacher then tests the gas with a	Brief the class: 'Goggles on. Two centimetres only. Describe what you SEE, not what you think should happen.' Circulate with a checklist and ask each group: 'What evidence tells you whether a reaction is happening?' WAIT TIME 10-15 seconds before prompting. At Station A, if learners report 'nothing', say: 'Is nothing a result? Look closely at the ribbon surface for tiny bubbles. Compare with the sodium video.' Before the demo say: 'Do not stare at the flame. Look sideways.' After the pop test ask 2 cold-called learners: 'What gas gives a pop with a lighted splint?' Ensure learners notice that Mg sinks in water	Structured observation table with 'evidence vs inference' columns: learners record only what they saw (bubbles, flame colour, white powder) in one column and what they think it means in another. This separates observation from interpretation and helps them build claims from evidence, which they will use in the Explain phase.	Observable indicators: (a) goggles on and 2 cm ribbon used by 100% of groups; (b) tables record at least 3 reactions with appropriate descriptors (no visible reaction in cold water; slow-to-steady bubbles in HCl; bright white flame in air); (c) at least 6 of 8 groups correctly name the pop-test gas as hydrogen; (d) groups distinguish observation from inference in their table. Teacher notes groups that write 'it reacted violently', which signals a Na-based assumption to correct.

	lighted splint (pop test) while learners watch. Demonstration C (air): Mr. Otieno holds a 3 cm Mg ribbon in tongs and ignites it behind a screen; learners watch the bright white flame from an angle and note the white powder (MgO) left behind. Each group fills an observation table with columns: test, what we saw, did it react (fast / slow / none). They also look at the Lesson 3 data table to recall Mg density 1.74 g/cm³ (sinks in water) and melting point 650 C, compared with Na 0.97 g/cm³ and 98 C.	(density 1.74 g/cm³) while Na floated. Remind the class that sodium, potassium and lithium are shown only by the teacher and are never handled by learners. Keep the first aid kit visible and name its location.		
Explain Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balancing Chemical Equations Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners work first in groups, then as a class. Step 1 (10 minutes): groups rank the reactions they saw from least to most vigorous and compare with the Lesson 4 sodium evidence (Na fast fizz and melts in water; Mg no visible reaction in cold water). Step 2: using the electron arrangements Na 2,8,1 and Mg 2,8,2, groups discuss: 'Both lose outer electrons, so why is Mg less reactive?' Guiding idea to build on: Mg has two outer electrons to lose, and the metal is denser and has a higher melting point (Lesson 3 data), so it holds on more strongly than Na and reacts less readily. Step 3: write and balance equations in their notebooks: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$; $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$; and compare with $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ and $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$. Learners count atoms on each side for every equation and show it in a small atom-count table. Step 4: place Na, Mg, Li, K, Ca in order of reactivity: $\text{K} > \text{Na} > \text{Li} > \text{Ca} > \text{Mg}$, justifying each position from the evidence. Step 5: relate to the phenomenon: Why is	Pose: 'Sodium and magnesium both lose outer electrons. Sodium fizzes in cold water, magnesium does not. What does that tell us about how easily each loses electrons?' WAIT TIME 10-15 seconds. Cold-call 4 learners to give a claim, then ask 2 others: 'Do you agree? What evidence supports that?' Model balancing $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ on the board: write the unbalanced $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$, count Mg and O on each side (1 Mg, 2 O on left; 1 Mg, 1 O on right), then add a coefficient 2 before MgO and then 2 before Mg. Then release groups to balance $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ themselves, checking H and Cl. Check understanding: 'If you change the small number in a formula, what happens to the substance? So what can we change?' (only coefficients). Address the Predict-phase misconception that all metals near sodium behave like sodium. Do not introduce ions, ionic bonding or electronegativity; keep the explanation at 'the metal atom loses its outer electrons'. Link	Claim-Evidence-Reasoning (CER) with a rank-and-justify task: learners state a claim (Mg is less reactive than Na), cite evidence (no reaction in cold water, slower bubbles in acid, higher melting point, two outer electrons), and give reasoning linking electrons to reactivity. Balancing equations with atom-count tables turns a procedure into a conservation-of-atoms argument.	Indicators: (a) at least 80% of groups write a CER with all three parts, citing at least two pieces of evidence; (b) learners correctly balance $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ and $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ with atom counts (exit slip of 3 items: balance $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$; state the number of outer electrons in Mg; order Na, Mg, K by reactivity); (c) learners correctly order $\text{K} > \text{Na} > \text{Li} > \text{Ca} > \text{Mg}$; (d) at least 70% correctly state that Mg has 2 outer electrons (2,8,2). Common errors to note: changing subscripts to balance; writing MgO_2; saying Mg is unreactive rather than less reactive than Na.

	Mg safe to handle as a ribbon while Na must be kept under paraffin oil? Which element would be unsuitable for a sufuria and why?	back: 'Why do we cook ugali in aluminium and not in Mg-rich or sodium pots? Think about reactivity and the evidence.'		
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Alkaline Earth Metals Practice Source: CK-12  Search ARES for similar readings	Akinyi and Kiprop, the DQB keepers, invite groups to revisit the board. Each learner (a) finds one earlier question that this lesson helps answer (for example Akinyi's 'Why is sodium dangerous?') and writes a short evidence note on a sticky note, and (b) writes one new question on a different-coloured sticky note (for example 'Why does Ca react less than Na but still fizz?', 'Is magnesium used in anything like the sufuria?', 'What makes chlorine different?'). Notes are sorted into the four categories: Properties, Reactivity, Uses and Safety. The class then chooses 2 new questions to follow up in Lessons 6 and 7.	Say: 'Walk to the DQB. Find a question we can now partly answer. Stick your evidence next to it, in your own words.' Give 2 minutes of silent writing (WAIT TIME 10-15 seconds before sharing). Cold-call 3 learners to read a new question aloud and ask the class: 'Which category does this belong in, and why?' Ask Kiprop to read one question about the yellow Thika Road lamps and signal that Lesson 6 will return to it. Move questions that are already answered to an 'Answered so far' strip, and mark those that need more evidence with a question mark. Praise questions that can be tested.	Driving Question Board as a living record of understanding: sorting questions into categories and attaching evidence shows learners how their knowledge grows and identifies gaps, keeping the investigation learner-driven and connected to the anchoring puzzle.	Indicators: (a) every learner adds at least one evidence note or new question; (b) at least 60% of new questions are testable and linked to reactivity, family position or use; (c) notes are placed in the correct category by at least 80% of learners; (d) at least 5 learners link their note to the outer-electron idea. Teacher photographs the DQB to track which questions remain open.
Model Building Phase  VIDEO: Visually understanding balancing chemical equations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Alkaline Earth Metals (Flexbook) Source: CK-12  Search ARES for similar	Kiprop and Akinyi lead the revision of the Class Family Model (wall periodic table). Learners annotate: (1) Group 2 column (Be, Mg, Ca) with 'two outer electrons' and 'less reactive than Group 1 in the same period'; (2) a note on Mg: burns with a bright white flame to form MgO, no visible reaction with cold water, reacts with dilute HCl giving H ₂ ; (3) a reactivity arrow K > Na > Li > Ca > Mg; (4) the two balanced equations. Each group then sketches a revised model on a sheet: Mg 2,8,2 and Na 2,8,1 shown as shell diagrams with the outer electrons circled, and a caption starting 'Magnesium is less reactive than sodium because...'. Finally learners write a 3-line answer to the lesson question: 'Is magnesium as dangerous as	Say: 'Our model last lesson said alkali metals are reactive because of one outer electron. What must we add or change after meeting magnesium?' WAIT TIME 10-15 seconds. Cold-call 3 learners for one revision each. Ask: 'What stays the same, what changes?' (family behaviour is linked to outer electrons, but the number and strength of holding matter, so position in the table predicts how reactive an element is). Push for the idea that position in the periodic table helps predict behaviour. Collect the revised models and exit slips. Remind learners of safety: no tasting or removing materials from the lab, wash hands, and dispose of residues as directed. Preview Lesson 6: halogens and noble	Cumulative model revision: learners add new evidence to a shared model and explicitly state what they are changing and why. Drawing shell diagrams and writing a 'because' caption connects particle-level structure to observed behaviour, supporting prediction of other elements' behaviour from their position.	Indicators: (a) at least 80% of revised models show Mg 2,8,2 with two circled outer electrons and Na 2,8,1 with one; (b) captions include a causal 'because'; (c) the 3-line answer states that Mg is less reactive than Na and gives evidence; (d) at least 60% of learners note that reactivity affects suitability for a job; (e) the class model is updated with the reactivity arrow and both balanced equations.

readings	sodium? What does that suggest about choosing an element for a job?' and list one open question about the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle.	gases, including chlorine in Mombasa water and neon signs.		
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners separate observation from inference at the stations? Which prediction misconception (all metals near sodium fizz in water) persisted and in how many groups? Did the safety routines (goggles, 2 cm ribbon, not looking at the flame) hold? How many learners could balance both equations unaided, and what did the errors reveal (changing subscripts, MgO₂)? Did the explanation stay at 'loses outer electrons' without drifting into ionic bonding? Which DQB questions should Lessons 6 and 7 target? What would you change in the grouping, timing or use of the Lesson 3 density and melting point data?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Magnesium ribbon showed no visible reaction in cold water, gave steady bubbles of hydrogen gas in dilute HCl (pop test positive), and burned with a bright white flame in air leaving white MgO powder. Sodium, by comparison, fizzes fast in water and melts.
What did I learn?	Alkaline earth metals (Group 2) have two outer electrons (Mg 2,8,2). Magnesium is less reactive than sodium. The reactivity order is K > Na > Li > Ca > Mg. Equations are balanced by adjusting coefficients so atoms are conserved.
How does this explain the phenomenon?	Both sodium and magnesium react by losing outer electrons, but magnesium reacts less readily, shown by no cold-water reaction and slower reaction with acid. Its position in the periodic table lets us predict this: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ and $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. So reactivity helps decide which element suits which job and how safely it can be handled.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Halogens and Noble Gases: Why Chlorine Cleans Mombasa Water and Neon Lights the Sign

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use evidence from chlorine's reactions (teacher demonstration, video and simulation) and noble gas data to explain how outer electrons make halogens reactive, why a tiny dose of chlorine makes Mombasa and Nairobi water safe, and why noble gases are unreactive and suited to signs and lighting.
Knowledge	By the end of the lesson, the learner should be able to: (a) state that halogens (Group 17) have 7 outer electrons and are reactive non-metals (Cl 2,8,7); (b) describe chlorine as a greenish-yellow, toxic gas that reacts with sodium: $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$; (c) explain that chlorine at about 0.5 mg/L kills the bacteria that cause cholera and typhoid, and that only a small safe dose is used; (d) state that noble gases (Group 18) have full outer shells (Ne 2,8; Ar 2,8,8; He 2) and are therefore unreactive; (e) outline the uses of neon in signs and relate the unreactive nature of noble gases to their applications.
Skills	Predicting from electron arrangements; interpreting a data table (boiling points, state, electrical conduction); developing and using models (annotating the class family model); constructing explanations from evidence; writing and balancing a chemical equation; communicating in group discussion (SEP: Analysing and interpreting data; Developing and using models; Constructing explanations).
Attitudes	Appreciate the uses of the elements of the periodic table in day-to-day life; show care for self and others by observing safety around toxic gases and reactive metals (value: Love); respect others' opinions during discussion (value: Respect); take responsibility for safe use of digital devices when watching the simulation (value: Responsibility). Core competencies: Critical thinking and problem solving; Communication and collaboration; Digital literacy; Self-efficacy. PCIs: Road safety (properties of elements that make them suitable for illuminating roads); Safety and security; Health (safe drinking water).
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lessons 4 and 5 showed that reactivity depends on outer electrons (alkali metals lose 1 easily; magnesium has 2 and is less reactive). Lesson 6 uses Group 17 and Group 18 to complete the picture: chlorine is reactive because it is one electron short of a full outer shell, yet a tiny dose makes Mombasa water safe, while noble gases with full shells are unreactive and so suit signs and lamps. This closes two threads from Lesson 1 (Mombasa chlorinated water and the street lamp) and prepares for Lesson 7 (transition metals) and Lesson 8 (choosing the right element for the job).
Safety Notes	Chlorine gas is toxic: learners do NOT handle chlorine. Mr. Otieno-style teacher demonstration only, in a fume cupboard or in a well-ventilated area outdoors behind a safety screen, using a very small amount; if no safe set-up is available, use the video and simulation instead. Teacher wears goggles and lab coat; learners stay at least 2 m back with goggles on. Sodium is never handled by learners and is kept under paraffin oil; keep water away from stored sodium. Anyone feeling throat irritation moves to fresh air immediately and the teacher uses the first aid kit. Dispose of residues as directed by the teacher. Never mix household bleach (e.g., Jik) with any acid or other cleaner.

B. LESSON OVERVIEW

Mama Wanjiru in Mombasa cooks ugali in an aluminium sufuria and asks: why is her tap water safe when chlorine as a gas is poisonous? Learners start by predicting what chlorine will do to sodium and why a lamp gas such as neon does nothing at all. They gather evidence from a teacher demonstration or video of chlorine (greenish-yellow gas, reacting with sodium to make salt, $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$), the water-treatment data (about 0.5 mg/L at the Mombasa water works), and a data table of Ne and

Ar electron arrangements and boiling points. They then explain reactivity in terms of outer electrons (Cl needs one more; Ne and Ar already have a full outer shell), post and sort new questions on the Driving Question Board, and annotate the class family model. This lesson connects the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle back to the periodic table: position predicts behaviour, and behaviour decides the job.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Noble Gases Practice Source: CK-12  Search ARES for similar readings	(About 12 min) Learners look at two items on the front bench: a clear bottle of Mombasa/Nairobi-style treated tap water and a photo of a neon sign next to a photo of a Thika Road sodium lamp. Individually, in their notebooks, they answer: (1) Chlorine atoms have 7 outer electrons (Cl 2,8,7). Predict what chlorine will do when it meets sodium (2,8,1), and draw a quick sketch of the outer electrons. (2) Neon has the arrangement 2,8 and argon 2,8,8. Predict whether neon will react with sodium, and why or why not. (3) Mama Wanjiru asks why chlorine gas is poisonous but her tap water is safe: write one idea. Learners then compare ideas with a partner and post one prediction on a sticky note.	Say: "Mama Wanjiru cooks ugali in her sufuria in Mombasa. She says chlorine is poisonous, so why is it in her tap water? Before I show you anything, write what you think." Hold up the electron arrangements on the board (Na 2,8,1; Cl 2,8,7; Ne 2,8; Ar 2,8,8). Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after each question before calling on anyone. Cold-call 4 learners (not volunteers) to share a prediction, and ask each: "What in the electron arrangement made you say that?" Do not say who is right. Record 3-4 predictions on the board under the headings 'Chlorine + sodium', 'Neon + sodium' and 'Safe water'. Remind learners of Lesson 4 (alkali metals lose their one outer electron easily) and Lesson 5 (Mg has two outer electrons and is less reactive).	Predict-Observe-Explain (commit to a prediction with reasoning): writing a prediction tied to electron arrangement makes learners' existing ideas about outer electrons visible and creates a reason to look at the evidence. Revisiting predictions later in the lesson lets them see their own thinking change.	Observe whether learners (a) refer to outer electrons rather than just 'chlorine is bad', (b) link sodium's behaviour from Lesson 4 to the new prediction, (c) connect 'full shell' to 'does not react' in the neon prediction. Tally on the board how many predict that neon reacts and how many that it does not. Note common ideas such as 'chlorine in water must be harmful' or 'all gases react' for use in the Explain phase.
Observe Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Noble Gases	(About 25 min) Learners gather evidence in three stations or segments and record in a three-column table (What I observed / What it shows about outer electrons / Link to the puzzle). Segment A (8 min): Teacher demonstration (or video) of chlorine: greenish-yellow gas in a gas jar; a small heated piece of sodium lowered into chlorine burns with a bright yellow flame, forming	For Segment A, stand behind the safety screen with goggles and lab coat; say: "Goggles on. Everyone stays behind the line. Chlorine is toxic, so I do the demonstration and you do the observing." If not safe to demonstrate, play the video and say: "Watch for three things: the colour of the gas, the flame, and the colour of the product." Ask: "What happened to the sodium? What is the white	Evidence table with data analysis (Analysing and interpreting data): learners separate what they saw from what it means and link each observation to outer electrons. Using numbers (0.5 mg/L, boiling points) turns the phenomenon into measurable evidence rather than opinion.	Check that learners (a) record at least three observations accurately (greenish-yellow gas, bright yellow flame, white solid NaCl); (b) balance $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ with 2 Na and 2 Cl on each side; (c) calculate 10 mg for 20 L and 500 mg for 1,000 L; (d) note that Ne and Ar are gases with very low boiling points that do not conduct or react. Use a thumbs-up/sideways check: 'Can

(Flexbook) Source: CK-12 Search ARES for similar readings	white solid NaCl. Learners note colour, smell warning (not smelled), flame and product, then copy the equation $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ and count atoms on both sides to check it balances. Segment B (8 min): In groups of 4, learners study a data card: Mombasa water works adds about 0.5 mg of free chlorine per litre, which kills the bacteria that cause cholera and typhoid; chlorine gas at high concentration is toxic, so only a small safe dose is used. They calculate how much chlorine is needed for a 20 litre jerrican ($0.5 \text{ mg/L} \times 20 \text{ L} = 10 \text{ mg}$) and for a 1,000 litre tank (500 mg), and discuss why dose matters. Segment C (9 min): Learners analyse a table: Ne colourless gas, 2,8, boils at -246°C, non-conductor; Ar colourless gas, 2,8,8, boils at -186°C, non-conductor; Cl_2 greenish-yellow gas, 2,8,7, boils at -34°C. They also use a short simulation (or video) of neon in a glass tube glowing red-orange when electricity passes through, and compare with the Thika Road sodium lamp's yellow-orange light. They note that neon and argon were not changed by the electricity and did not react with anything.	solid? Where have we met it?" WAIT TIME of 10-15 seconds before calling on a learner; cold-call 3 learners. Prompt: "Sodium lost its one outer electron in Lesson 4. Where might that electron have gone?" (without formal ionic bonding detail: say only that the sodium atom gives up its outer electron and chlorine takes one). For Segment B, circulate and ask each group: "If 0.5 mg/L is safe and kills bacteria, what would happen if the dose were 100 times bigger?" For Segment C, ask: "What pattern do you see between the electron arrangements of Ne and Ar and their behaviour?" Cold-call 4 groups to share one pattern each. Check that groups compute 10 mg and 500 mg correctly and that the equation balances (2 Na, 2 Cl on each side).		you say in one sentence why chlorine is reactive?' and note learners who need support.
Explain Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Noble Gases	(About 20 min) In groups of 4, learners build a written explanation using a claim-evidence-reasoning frame to answer two linked questions: (1) Why is chlorine reactive, and why can a tiny dose of 0.5 mg/L make Mombasa water safe even though chlorine gas is toxic? (2) Why are neon and argon unreactive, and why does that suit them for signs and lamps? They must use: Cl 2,8,7 (7 outer	Say: "Use the frame: Claim, Evidence, Reasoning. Your reasoning must mention outer electrons." Provide the sentence starters: "Chlorine is reactive because its atoms have ___ outer electrons, so they ___." and "Neon is unreactive because ___." Circulate and ask: "Why does the same chlorine that kills bacteria at 0.5 mg/L harm people at high concentration?" and "Is a gas that	Claim-Evidence-Reasoning with gallery walk (Constructing explanations): learners must link observations and numbers to the mechanism (outer electrons), so the explanation is evidence-based rather than a memorised statement. Peer critique during the gallery walk sharpens reasoning.	Look for explanations that (a) state Cl has 7 outer electrons and is one short of a full shell; (b) state Ne (2,8) and Ar (2,8,8) have full outer shells so do not react; (c) use 0.5 mg/L to explain why only a small dose is used; (d) connect NaCl to the salt that is safe to eat. Use a 3-2-1 rubric: 3 = all four elements, 2 = two or three, 1 = one or none. Record names of learners with misconceptions for follow-up and

Practice Source: CK-12 Search ARES for similar readings	electrons, one short of a full shell of 8); Na 2,8,1 (one outer electron easily lost); Ne 2,8 and Ar 2,8,8 (full outer shells, nothing to gain or lose); the equation $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$; and the dose data. They also explain the sodium-lamp puzzle: the Thika Road sodium vapour lamp gives yellow light, neon signs glow red-orange, and why both are chosen for the job (neon is unreactive and so safe and stable inside a sealed glass tube; sodium is used as a vapour for its yellow light). Groups then revisit their predictions from the Predict phase, mark each as supported or not supported, and correct it. Finally, they use the gallery walk: groups swap explanations and give one star and one question.	does not react the same as a gas that is useless? Where in Kenya do we use it?" WAIT TIME of 10-15 seconds after each question. Cold-call 5 learners from different groups to read a claim aloud, and ask a different group to add evidence. Address misconceptions: (i) 'chlorine in tap water is the same as chlorine gas leaking' (clarify dose: 0.5 mg/L is a tiny safe amount); (ii) 'noble gases are unreactive because they are cold' (clarify full outer shell); (iii) 'the sodium lamp contains neon' (sodium vapour lamp uses sodium). Do not introduce ionic bonding detail, ion formulas or electronegativity: say only that the sodium atom gives up its outer electron and the chlorine atom takes one, giving the stable compound NaCl (salt). Remind them that salt (NaCl) is safe to eat even though sodium and chlorine alone are dangerous.		plan a short re-teach with the electron arrangements on the board.
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Noble Gases (Flexbook) Source: CK-12 Search ARES for similar readings	(About 10 min) Akinyi and Kiprop, the DQB keepers, lead the class to revisit the Driving Question Board. Learners (1) pick at least one earlier sticky note they can now answer (for example Kiprop's 'Why do street lamps glow yellow?' or 'Why is chlorine in our water?') and write the answer on the back citing evidence from this lesson; (2) write one new question on a new sticky note (for example 'Why does iron rust but gold does not?', 'Why is the sodium lamp yellow and the neon sign red?', 'Is chlorine in water safe for everyone?', 'Which element is best for a new sufuria?'); and (3) place each note under 'Properties', 'Reactivity', 'Uses' or 'Safety'. The class counts how many 'Reactivity' and 'Safety'	Say: "Akinyi, Kiprop, which questions on our wall can we now answer using chlorine and the noble gases?" Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before sharing. Cold-call 4 learners to read a note they answered and state the evidence used. Ask: "Which question can we NOT answer yet? What evidence would we need?" Redirect questions about transition metals, hardness and rusting to Lesson 7. Praise good questions that link position in the periodic table to a job. Keep the DQB visible and move answered notes into a 'Now we can explain' column.	Driving Question Board (tracking growing understanding): linking old questions to new evidence shows learners their own progress, and new questions keep the inquiry going toward the next lesson.	Check that (a) each learner posts one answer with evidence and one new question; (b) sorted notes are in the right category; (c) at least two questions are now marked answered (for example about the Mombasa water and the street lamp). Observe whether new questions move from 'what happens' to 'why it happens', and note learners still asking only descriptive questions.

	questions are answered and agrees which questions the next lesson (transition metals) should tackle.			
Model Building Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Noble Gases Practice Source: CK-12  Search ARES for similar readings	(About 13 min) At the class wall periodic table (the Family Model), learners add this lesson's evidence. In groups they annotate Group 17 (F, Cl): '7 outer electrons, one short of a full shell, reactive non-metals; Cl is a greenish-yellow toxic gas used at about 0.5 mg/L to treat Mombasa and Nairobi water; $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ '. They annotate Group 18 (He, Ne, Ar): 'full outer shell (He 2, Ne 2,8, Ar 2,8,8), unreactive, gases with very low boiling points; Ne used in signs, Ar in lamps'. They then update the model's rule: 'Reactivity depends on outer electrons: Group 1 lose 1 easily (very reactive), Group 2 lose 2 (less reactive), Group 17 are one short of a full shell (reactive non-metals), Group 18 are full (unreactive).' Each learner draws a quick sketch of Na, Cl and Ne showing their outer electrons and adds a one-line note on the job each suits. They finish by completing the summary table (below) in their notebooks.	Say: "Our family model must predict, not just describe. Using the position of an element in the table, what can we predict?" Hand a marker to a different group for each annotation. WAIT TIME of 10-15 seconds before groups propose wording. Cold-call 4 learners to defend a line on the model with evidence. Ask: "How does the model now explain the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle? Which parts are still missing?" Point out that Lesson 7 will add transition metals (copper wire, iron and zinc for mabati) and Lesson 8 will use the whole model to choose the right element for a job. Check that every annotation uses 'outer electrons' (not 'valence electrons'), 'chemical family' and 'group', and that no ionic bonding detail has been added.	Cumulative model revision (Developing and using models): adding new evidence to the shared class periodic table and refining the rule linking outer electrons to reactivity and use turns separate facts into one predictive model that can be applied to the driving question.	Check that each learner's sketch shows Na 2,8,1, Cl 2,8,7 and Ne 2,8 correctly and that annotations on the model name the family, the number of outer electrons, the behaviour and one use. Exit check: ask 3 cold-called learners to state, in one sentence each, why chlorine is used in small doses and why neon suits signs. Collect the summary tables to gauge who needs support before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners explain reactivity using outer electrons (Cl one short of a full shell; Ne and Ar already full) rather than saying 'chlorine is bad' or 'noble gases are lazy'? (2) Could they use the 0.5 mg/L data to explain why the dose matters and why a toxic gas can make water safe? (3) Did any learners think the sodium lamp contains neon, or that salt is dangerous because sodium and chlorine are? How did you address it? (4) Was the chlorine demonstration safe and well ventilated, or did you rely on the video, and what would you change? (5) Which DQB questions remain unanswered and how will Lesson 7 on transition metals address them? (6) Which learners need extra help balancing $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ or drawing electron arrangements, and what support will you provide?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Record what you observed about chlorine (colour, reaction with sodium, product NaCl), the water treatment data (about 0.5 mg/L of chlorine, 10 mg for 20 L), and the data for neon and argon (colourless gases, 2,8 and 2,8,8, boiling points -246 C and -186 C, unreactive, non-conductors).
What did I learn?	Write what you learned: halogens (Group 17) have 7 outer electrons and are reactive non-metals; chlorine kills the bacteria that cause cholera and typhoid at a tiny dose, but is toxic in large amounts; noble gases (Group 18) have full outer shells (He 2, Ne 2,8, Ar 2,8,8) and are unreactive; the balanced equation $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$.
How does this explain the phenomenon?	Explain, using outer electrons, why Mombasa tap water treated with chlorine is safe for Mama Wanjiru's ugali, why sodium and chlorine make safe table salt together, and why unreactive noble gases such as neon suit signs and road lighting. State how this helps answer the driving question: how the periodic table helps us match an element to a job.

LESSON 7 (80 minutes (2 x 40-minute lessons)): Transition Metals: Matching Properties to Jobs (Copper, Iron, Zinc, Lead and Aluminium)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners use physical property data to explain why transition metals (and lead and aluminium for comparison) are chosen for particular jobs in Kenyan daily life, and to weigh trade-offs such as the toxicity of lead.
Knowledge	By the end of the lesson, the learner should be able to: (a) locate the transition block on the periodic table (between Group 2 and Group 13, from Period 4) and state that Fe (Z=26), Cu (Z=29) and Zn (Z=30) are transition metals; (b) state that transition metals are generally strong, dense, good conductors with high melting points, using the data (Fe 7.87 g/cm ³ , 1538 C; Cu 8.96 g/cm ³ , 1085 C; Zn 7.14 g/cm ³ , 420 C); (c) outline the uses of Cu (electrical wire), Fe (structural steel and mabati base), Zn (galvanising coat that protects iron by sacrificial protection), Pb (car batteries and radiation shielding) and Al (sufuria); (d) explain why lead is being phased out despite its usefulness, because it is dense but toxic.
Skills	Developing and using models (matching table); analysing and interpreting data (comparing density, melting point and conductivity); obtaining, evaluating and communicating information from digital and print sources; constructing explanations and arguing from evidence (claim-evidence-reasoning for an element-to-job match); observing and comparing samples of Fe, Cu, Zn and Al safely.
Attitudes	Appreciate the uses of elements of the periodic table in day-to-day life (KICD Sub-Strand 2.2); respect peers' opinions during discussion; take care of self and others by observing safety (Love); use digital devices responsibly (Responsibility). Core competencies: Digital literacy as the learner uses available digital and print media to research on transition elements and their uses. PCI: Safety and security (handling metal samples and lead information).
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lessons 4 to 6 explained reactive families (alkali metals, magnesium, halogens and noble gases) using outer electrons. This lesson turns to the strong, dense metals we meet daily in the copper wire and the zinc-coated iron mabati of Lesson 1, and gives the class evidence that a job is matched to specific properties, with trade-offs. It builds on Lesson 2 (position of the transition block) and Lesson 3 (physical properties data), and prepares Lesson 8, where learners choose the right element for a new Kenyan product.
Safety Notes	Learners may handle small samples of Fe, Cu, Zn and Al with goggles on; no sodium, potassium or lithium is handled by learners. Lead is NOT handled by learners; it is studied only through data, pictures and video because it is toxic. Wash hands after handling metal samples; do not put samples near the mouth; wear goggles; tell the teacher about cuts from sharp edges and use the first aid kit. Handle digital devices appropriately and follow the school's online safety rules.

B. LESSON OVERVIEW

Mama Wanjiru's aluminium sufuria, the copper wire in the kettle cable and the zinc-coated iron mabati from Kericho farm buildings are all metals, yet each does a different job. Learners predict which metal suits which job, then gather evidence by examining Fe, Cu, Zn and Al samples and a physical data table (Fe, Cu, Zn, Pb, Al), and by researching uses with print and digital sources. In the Explain phase they build claim-evidence-reasoning arguments matching property to use, including why zinc protects iron and why lead is dense but toxic so its use is being reduced. They then post new questions on the DQB and add the transition block and uses to the class family model, advancing the driving question of how the periodic table helps match element to job.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Transition Metals (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners look at four anchor objects or pictures from Lesson 1: the aluminium sufuria, the copper wire from a kettle cable, a piece of mabati (iron coated with zinc) and a picture of a car battery (lead). On a prediction sheet they individually write: (1) which property of each metal makes it right for its job, and (2) what they think would go wrong if the metals were swapped (e.g. a lead cooking pot, an aluminium mabati roof, a zinc electric wire). They then share with a partner and keep the sheet to revisit in the Explain phase.	Hold up the sufuria and say: "Mama Wanjiru in Mombasa cooks ugali in this. Why this metal and not the metal in the mabati roof? Write your prediction first, silently." Give WAIT TIME of 10-15 seconds before any talk and 2 minutes of silent writing. Then say: "Turn to your partner and compare. Do you agree on at least one property?" Cold-call 4 learners from different parts of the room, accepting all ideas without judging, and record keywords (strong, light, conducts, does not rust) on the board. Ask: "What would you want to see in the data to test that idea?" Point to the Lesson 2 wall periodic table and ask: "Where do Fe, Cu and Zn sit? Come and point." Invite 2 learners to show the block between Group 2 and Group 13 starting at Period 4. Remind them that this is the position taught in Lesson 2.	Prediction with elicitation of prior ideas: writing a prediction before seeing data makes learners commit to an initial model of 'property to job', so that later evidence can confirm or revise it. Linking the objects to the family model keeps the phenomenon (the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle) in view.	Scan prediction sheets and look for: (a) at least one named property per metal (conductivity, strength, low mass, rust protection); (b) whether learners mention reactivity ideas from Lessons 4 to 6 inappropriately; (c) whether at least 3 learners can correctly point to the transition block. Tally common ideas (e.g. 'metals are all strong') on the board to revisit as misconceptions.
Observe Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Transition Metals (Flexbook) Source: CK-12  Search ARES	(25 min) In groups of 4, learners rotate through two stations for 10 minutes each, then spend 5 minutes on a research task. Station 1 (samples, goggles on): examine and compare small samples of Fe (nail), Cu (wire), Zn (strip or galvanised nail) and Al (offcut from a sufuria): colour, shine, how easily each bends, mass of equal-sized pieces by feel or balance, and whether each conducts using a simple cell, bulb and wires. They record results in a table. Station 2 (data): they study the physical data table for Fe, Cu, Zn, Pb and Al: density (Fe 7.87, Cu 8.96, Zn 7.14, Pb 11.34, Al	Before the rotation say: "Goggles on. Samples are for looking, bending and testing only; nothing goes near your mouth. Lead is not in the box. We study it from data only because it is toxic." Circulate and ask: "Which sample felt heavier for the same size? What does the density table say?" and "What did the bulb do for each metal? Is that what the data says about Cu?" Pause with WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report the densest and the highest-melting metal. Ask: "Fe, Cu and Zn are all transition metals. What do they have in common in the data	Data table analysis with pattern finding (Gallery of evidence): comparing hands-on observations with numerical data lets learners see patterns across metals (high density and melting points for transition metals; low density for Al; low melting point for Pb and Zn). Linking back to Lesson 3's method for comparing properties reinforces data-based reasoning.	Check that groups: (a) correctly record at least 3 observations per sample; (b) read the data table correctly, naming Pb as densest (11.34 g/cm ³), Fe as highest melting (1538 C) and Pb as lowest melting among Fe, Cu, Zn, Pb, Al (327 C); (c) observe goggles and handling rules; (d) correctly pair each metal with its use and give one reason. Note groups that confuse density with strength or that claim all metals conduct equally.

for similar readings	2.70 g/cm³), melting point (Fe 1538, Cu 1085, Zn 420, Pb 327, Al 660 C) and conductivity (Cu excellent; Fe, Zn, Al conduct; Pb poor-to-moderate). They answer: which is densest, which has the highest melting point, which has the lowest melting point among the five, and which is lightest. Research task: using print resources or a digital device (offline resources if no internet), each group finds the uses of one metal (Cu - wiring, Fe - steel and mabati base, Zn - galvanising, Pb - car batteries and radiation shielding, Al - sufuria) and notes one reason for the use.	compared with Na and K from Lesson 3?" (higher density, higher melting point, harder; Na and K float, these sink). Prompt: "Aluminium is not in the transition block. Does it still fit a job? Why?" Assign each group one metal for the research task and give 5 minutes. Have first aid kit visible.		
Explain Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Transition Metals Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	(25 min) Learners return to their predictions and write one claim-evidence-reasoning (CER) paragraph for the element-to-job match assigned by the teacher: (1) Cu for the kettle wire, (2) Fe for the mabati base and steel, (3) Zn for the coat on mabati, (4) Al for the sufuria, (5) Pb for car batteries. They use data: e.g. Cu has excellent conductivity and is ductile so can be drawn into wire; Al has low density (2.70 g/cm³), heats food quickly and resists corrosion through its oxide layer; Fe is strong and cheap but rusts, so it is coated with Zn, which is more reactive than iron and corrodes first (sacrificial protection), which links to Lesson 5's idea that more reactive metals lose electrons more readily and to the $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ reaction. For lead, groups discuss the trade-off: Pb is very dense (11.34 g/cm³), good for batteries and radiation shielding, but toxic, so its use is being reduced and never suits a cooking pot. Groups	Model one CER: "Claim: copper suits wire. Evidence: its conductivity is excellent and it is ductile. Reasoning: current must pass easily and the metal must be drawn into thin strands." Then give 4 minutes for groups to write their own. Ask: "Mama Wanjiru's pot is safe but she asks why chlorine gas is poisonous. For metals, what makes lead a poor choice for a pot even though it is soft and melts at 327 C?" Use WAIT TIME of 10-15 seconds, then cold-call 3 learners. Ask: "If a mabati sheet is scratched, why does the iron underneath still not rust straight away?" Guide them to sacrificial protection: zinc is more reactive, so it is used up first. Use the sentence stem: "The metal is chosen because ____ (property), shown by ____ (data), but the trade-off is ____." Correct the misconception that all transition metals are equally good conductors by returning to the data table. During the job interview, cold-call 5 different groups and	Claim-Evidence-Reasoning (CER) with structured argumentation: learners must link a property in the data to a job and name the trade-off, building the habit of justifying choices with evidence. The job interview applies the matching model to new contexts, showing why the periodic table and properties together help predict suitable elements.	Look for CER paragraphs that: (a) state a clear claim; (b) quote at least one correct number from the data (e.g. Al 2.70 g/cm³, Pb 11.34 g/cm³, Zn 420 C); (c) give a scientific reason; (d) name a trade-off such as lead toxicity, rusting of iron or cost. Use a 3-point check (claim, evidence, reasoning). Listen for correct use of 'sacrificial protection' and 'zinc more reactive than iron'. Note learners who suggest lead for cooking or copper as a roof without addressing cost.

	then hold a whole-class 'job interview': a learner reads a job (e.g. 'overhead cable in Nairobi', 'cooking pot for Gikomba jua kali artisans', 'roof in Kericho') and groups argue which metal to hire and why. Learners then revise their original predictions in a different colour.	invite counter-arguments with: "Who can respectfully challenge that with a different data point?" Remind learners not to introduce bonding detail, since it is not part of this lesson.		
Driving Question Board (DQB) Creation  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Transition Metals (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min, shared with the end of Explain) Akinyi and Kiprop, the DQB keepers, lead the class in adding new questions on sticky notes (e.g. 'Why is copper cable so much more expensive than iron?', 'What happens to old car batteries in Nairobi?', 'Why is aluminium not a transition metal but still used for sufuria?'). Learners sort the notes under Properties, Reactivity, Uses and Safety, and move any earlier questions they can now answer (e.g. 'Why is the sufuria safe?', 'Why does mabati not rust quickly?') to an 'Answered' column with a short evidence note.	Say: "Look back at our board from Lesson 1. Which question can we now answer with evidence from today?" Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to move a note and justify it with data. Ask each learner to write one new question on a sticky note and hand it to Akinyi or Kiprop to place. Say: "Which question needs evidence we do not have yet? That will help us in Lesson 8." Point out gaps (cost and availability) as items to weigh in Lesson 8. Praise respectful listening.	Driving Question Board tracking: moving answered questions and posting new ones makes learning progress visible, links the lesson back to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle and shows how the evidence advances the driving question.	Count how many learners contribute a new question and whether notes are placed in the correct category. Check that at least 3 learners move a note to 'Answered' with a data-based reason. Look for questions that show curiosity about cost, safety or recycling, which signal readiness for Lesson 8.
Model Building Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Transition Metals	(10 min) Learners add to the class Family Model (wall periodic table): they mark the transition block (Sc to Zn, including Fe ₂₆ , Cu ₂₉ and Zn ₃₀) and place labelled cards for Cu - wiring (excellent conductor, ductile), Fe - steel and mabati base (strong, cheap), Zn - galvanising (more reactive than iron, sacrificial protection), Al - sufuria (light, conducts heat, oxide layer), and Pb (Z=82, Group 14, a post-transition metal taught here as a heavy metal) - batteries and shielding (dense, toxic). In their notebooks they add a mini table:	Say: "Come and place one card on the model and say aloud the property and the data that support it." Invite 5 learners (cold-call, different groups) and ask the class to agree or correct with a thumbs signal. Clarify: "Pb is in Group 14, not the transition block, and Al is in Group 13, but we compare them here for use." Ask: "How does the periodic table help predict which metal could do a job? What else besides position do we need?" (data on properties, safety, cost, availability). Use WAIT TIME of 10-15 seconds before each cold call.	Cumulative model revision: adding transition metals and their uses to the wall periodic table extends the class model from reactivity (Lessons 4 to 6) to properties and uses, helping learners see how position, properties and trade-offs combine to answer the driving question.	Check that learners place Fe, Cu and Zn in the transition block and correctly label Pb and Al as outside it. In notebooks, look for a complete mini table with one correct number, one use and one trade-off for each of 5 metals. Ask 3 learners to give a one-sentence answer to 'Why is copper used for wire and lead not used for pots?' and record accuracy.

(Flexbook) Source: CK-12 Search ARES for similar readings	metal, key property with number, use, trade-off. They also add a line: 'Transition metals are strong, dense, good conductors with high melting points, so uses are matched to specific properties.'	Close by connecting to the earlier families: "Reactive families like sodium are dangerous alone; these metals are chosen for strength, conduction and low reactivity." Preview Lesson 8: choosing the right element for a new Kenyan product.		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners use numbers from the data table (not just general statements) in their CER paragraphs? (2) Did any learners confuse density with strength, or assume lead and aluminium are transition metals? (3) Was the sacrificial protection idea understood by linking to zinc's higher reactivity? (4) Did every learner participate safely with the metal samples and avoid contact with lead? (5) Which DQB questions about cost, recycling or availability should be addressed in Lesson 8? Note which learners need support with reading data tables or writing reasoning, and adjust groups. Check that offline print resources were sufficient where digital devices were limited.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We examined samples of Fe, Cu, Zn and Al and a data table for Fe, Cu, Zn, Pb and Al: density (Fe 7.87, Cu 8.96, Zn 7.14, Pb 11.34, Al 2.70 g/cm ³), melting point (Fe 1538, Cu 1085, Zn 420, Pb 327, Al 660 C) and conductivity (Cu excellent). We also researched their uses: wiring, mabati, galvanising, batteries and sufuria.
What did I learn?	Transition metals such as Fe, Cu and Zn are generally strong, dense, good conductors with high melting points, and each use matches a specific property. Fe is strong and cheap, Cu is an excellent conductor and ductile, Zn coats iron and protects it by sacrificial protection, Al is light and resists corrosion, and Pb is very dense but toxic so its use is being reduced.
How does this explain the phenomenon?	Mama Wanjiru's sufuria is aluminium because it is light, conducts heat and resists corrosion; the kettle wire is copper because it conducts electricity excellently and can be drawn into wire; the mabati roof is iron for strength coated with zinc because zinc is more reactive and corrodes first. Elements are chosen for properties, balanced against trade-offs such as lead's toxicity, cost and availability.

LESSON 8 (80 minutes (2 x 40-minute lessons)): Putting It Together: Choosing the Right Element for the Job

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use evidence from properties, reactivity and periodic table position to justify choosing an element for a new Kenyan product, weighing safety, cost and availability, and complete the family model and the Driving Question Board.
Knowledge	By the end of the Sub Strand, the learner should be able to: a) identify chemical families in the periodic table, b) classify elements of the periodic table into their chemical families, c) investigate the physical and chemical properties of selected elements of the periodic table, d) outline the uses of elements of the periodic table, e) appreciate the uses of the elements of the periodic table in day-to-day life. In this lesson learners link all of these: elements are chosen for jobs because of their properties, which depend on outer electrons and so on position in the periodic table. Sodium (2,8,1) reacts violently with water because its single outer electron is lost very easily, but combined with chlorine as NaCl it is stable and safe to eat. Chlorine kills microbes at about 0.5 mg/L in Mombasa water. Aluminium suits the sufuria, copper suits wire, zinc coats iron for mabati, and neon suits signs.
Skills	Science and Engineering Practices: constructing explanations and designing solutions (final explanation and product recommendation); engaging in argument from evidence (defending an element choice with a claim, evidence and reasoning); developing and using models (comparing the Lesson 1 model with the final family model); obtaining, evaluating and communicating information. Core competencies: Self-efficacy and Digital literacy (research and present transition metal uses), Critical thinking and problem solving.
Attitudes	Love: take care of self and others by observing safety when handling elements and appropriately using first aid kits in case of injuries. Respect: listen to and consider peers' arguments. Appreciate the uses of the elements of the periodic table in day-to-day life. PCI: Road safety (properties that make sodium suited to illuminating roads) and Safety and security.
Key Inquiry Question	How are elements suited to their use in day-to-day life?
Purpose in Storyline	Lesson 8 is the culmination. Learners return to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle from Lesson 1 and answer the Driving Question Board using the evidence from Lessons 1-7: families and layout (L2), physical property patterns (L3), alkali metal reactivity and outer electrons (L4), magnesium and reactivity order (L5), halogens and noble gases (L6) and transition metal uses and trade-offs (L7). They write a final explanation and recommend an element for a new Kenyan product.
Safety Notes	No new reactive-metal practical is done in this lesson. If the sodium demonstration is replayed, only Mr. Otieno-style teacher demonstration behind a safety screen with goggles and lab coat, a pea-sized piece (about 0.1 g) only, sodium kept in paraffin oil, no water near stored sodium. Learners do not handle Na, K or Li. Learners may handle Mg ribbon, Al, Cu, Fe and Zn samples with goggles on. Keep the first aid kit visible. Remind learners that lead is toxic and samples must be handled with clean hands washed afterwards, and chlorine gas is never produced in class.

B. LESSON OVERVIEW

In Lesson 1 learners met a puzzle: the aluminium sufuria, copper wire, mabati roof (iron coated with zinc), table salt, chlorinated Mombasa water and the Thika Road sodium lamp are all useful and safe, yet sodium reacts violently with water and chlorine gas is poisonous. Mama Wanjiru, a food seller in Mombasa, asked why her pot is safe while chlorine in her tap water is poisonous as a gas. In this final lesson learners predict which element they would choose for new products, revisit the evidence cards from Lessons 1-7, build a final explanation using claim-evidence-reasoning, complete the DQB by sorting answers under Properties, Reactivity, Uses and Safety, and compare their Lesson 1 initial model with the final class family model. They finish by recommending an element for a new Kenyan product, weighing properties, safety, cost and availability, and reflect on caring for self and others.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown a design challenge card: 'A jua kali artisan in Gikomba wants to make (a) a cooking pot, (b) a cable for a charger, (c) a roof sheet for a farm store in Kericho. Which element would you pick for each: sodium, aluminium, copper, iron coated with zinc, or neon?' Individually, in 3 minutes, learners write a prediction and one reason for each, using only what they already know. They then also predict: 'Which of these elements could we NOT use on its own, and why?' They record predictions in their science notebooks to compare with the final explanation later.	Display the six anchor objects (sufuria, copper wire, mabati, salt, a bottle of treated water, picture of the Thika Road lamp) and the challenge card. Say: 'Remember Mama Wanjiru's question? Before we revisit our evidence, write what you think. There are no wrong predictions today.' WAIT TIME of 10-15 seconds after posing each question before anyone speaks. Cold-call 4 learners (not volunteers) to share one choice and one reason; ask a fifth: 'Who disagrees with that choice and why?' Record two competing predictions on the board without judging. Ask: 'Which prediction would you most like us to test with evidence?'	Eliciting prior ideas through individual written prediction then cold-call sharing. Writing predictions before revisiting evidence makes learners' initial thinking visible, creates cognitive conflict between peers' choices, and gives a baseline to compare with their final explanation, so they see their own learning growth.	Observe whether learners (1) name properties such as light, conductive or unreactive in their reasons, (2) mention safety for sodium, (3) refer to the periodic table or outer electrons. Tally on a checklist: how many give property-only reasons, how many mention family or outer electrons, how many choose sodium for a pot. Use the tally to decide which evidence card to emphasise in the Observe phase.
Observe Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners do an evidence gallery review. Each group receives a set of data cards drawn from the lesson sequence: (1) Physical data table: Na density 0.97 g/cm ³ , melting 98 C, soft; Al 2.70 g/cm ³ , melting 660 C; Cu 8.96 g/cm ³ , melting 1085 C, excellent conductor; Fe 7.87 g/cm ³ , melting 1538 C; Zn 7.14 g/cm ³ , melting 420 C; Pb 11.34 g/cm ³ , melting 327 C, toxic; Ne boils at -246 C, Ar at -186 C. (2) Reactivity order K > Na > Li > Ca > Mg with 2Na + 2H ₂ O -> 2NaOH + H ₂ and 2Mg + O ₂ -> 2MgO. (3) Electron arrangements: Na 2,8,1; Mg 2,8,2; Cl 2,8,7; Ne 2,8; Ar 2,8,8. (4) Chlorine data: greenish-yellow gas, 0.5 mg/L in Mombasa water, 2Na + Cl ₂ -> 2NaCl. (5)	Say: 'Your job is to become the evidence experts. Every claim you make today must be backed by a card.' Circulate and ask each group: 'Which card tells you why sodium cannot be a sufuria? Show me.' and 'What does the 2,8,1 arrangement tell you about how sodium behaves?' WAIT TIME 10-15 seconds before prompting. Check goggles are on when samples are handled. Cold-call 3 groups to report one row of their matrix. Prompt: 'Where does Al sit compared with the four families we studied? What does that tell us about naming it a transition metal or not?' (Al is an other metal, not in the four families.) Correct misconceptions: salt (NaCl) is a compound and is not the same as	Evidence-card gallery with an Element Job Matrix (data organisation and pattern finding). Structuring data in one table across families lets learners see that properties come from outer electrons and position, and that safety depends on whether an element is free or combined and on dose.	Check each matrix for: correct family for Na, Mg, Cl, Ne, Cu, Zn; correct outer electrons (Na 1, Mg 2, Cl 7, Ne 8); a property tied to a use for each of Al, Cu, Fe, Zn; at least one trade-off (Pb toxic, Na reactive). Success indicator: 80% of groups complete 5 rows correctly. Note groups who put Al in a family or confuse NaCl with Na for reteaching.

	Uses list: Cu wire, Fe structural steel and mabati base, Zn galvanising by sacrificial protection, Pb batteries but toxic, Al sufuria with oxide layer. Learners also handle real samples (Mg ribbon, Al, Cu, Fe, Zn) with goggles on and compare appearance and feel. Each group fills an Element Job Matrix with columns: element, family and position, outer electrons, key property, safe use, trade-off. They also locate each element on the wall family model.	sodium; chlorine gas and chlorine in 0.5 mg/L water are the same element at very different doses.		
Explain Phase  VIDEO: Electronegativity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners write the Final Explanation individually using a claim-evidence-reasoning frame to answer the Driving Question and Mama Wanjiru's question: 'Why is sodium dangerous alone but salt safe, why is chlorine poisonous as a gas but safe in Mombasa water, and why is each element chosen for its job?' Frame: Claim (an element is chosen for its job because...), Evidence (at least three data points from cards, e.g. Na 2,8,1 loses its outer electron easily, $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$; 0.5 mg/L chlorine kills cholera and typhoid bacteria; Al is light 2.70 g/cm³ with an oxide layer; Cu excellent conductor), Reasoning (links to outer electrons and periodic table position, and to safety, cost, availability). Then pairs swap and use a checklist to give one star and one wish. Next, groups design a new product recommendation: choose one scenario (water pipe for a Kisumu school, cooking pot for Gikomba jua kali, electric cable, bright sign for a Nairobi shop, roofing for a Kericho tea store) and argue for an element, rejecting at least one	Model the first sentence of the frame: 'Sodium is unsuitable for a sufuria because its one outer electron is lost very easily, so it reacts violently with water.' Then say: 'Now it is your turn. Use at least three pieces of evidence and one trade-off.' WAIT TIME 10-15 seconds after each prompt. Questions to push reasoning: 'Why does combining with chlorine make sodium safe?' (the sodium atom has lost its outer electron and the compound NaCl is stable; do not go into ionic bonding detail.) 'If copper is such a good conductor, why not use it for the sufuria?' (cost, availability; Al is light and cheaper.) 'Why do we put zinc on iron?' (sacrificial protection). Cold-call 5 learners to read their reasoning sentence aloud and ask peers to name the evidence used. For each group's product pitch, ask: 'What is the safety risk and what is the cost trade-off?' Allow 90 seconds per group pitch and have two groups challenge each pitch.	Claim-Evidence-Reasoning argumentation with peer critique. Requiring each claim to be tied to data and to atomic structure builds an explanation that links the macroscopic uses with the particle-level cause, and the product-pitch design task transfers the explanation to a new Kenyan context, which tests true understanding.	Rubric on the final explanation (4 levels): Exceeds, Meets, Approaches, Below. Meets requires a correct claim, three data points, reasoning linking outer electrons or position to behaviour, and one safety or cost trade-off. Observe whether learners distinguish element from compound, and free chlorine gas from a 0.5 mg/L dose. Collect explanations; target feedback on those who cite uses without referring to outer electrons or periodic table position.

	alternative with evidence.			
Driving Question Board (DQB) Creation  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Akinyi and Kiprop lead the class back to the Driving Question Board with sticky notes sorted into Properties, Reactivity, Uses and Safety. Each group takes 4 original questions from Lesson 1, such as Akinyi's 'Why is sodium dangerous?' and Kiprop's 'Why do street lamps glow yellow?', and writes an answer on a new coloured note using evidence and the lesson where it was learned. Learners mark each question Answered, Partly answered or New question. They add at least one new question for further inquiry, for example 'Why do other metals give different flame colours?' or 'How is lead replaced in batteries?'. The class counts how many questions moved from open to answered in each category.	Say: 'Akinyi and Kiprop, read your first questions aloud. Who can answer one with evidence?' WAIT TIME 10-15 seconds. Cold-call 4 learners to place an answer note and justify it with a card. Ask: 'Which question can we only partly answer? What evidence would we need?' Ensure each category has at least two answers. Redirect off-topic or incorrect answers by asking 'Which evidence card supports that?' Remind: the Thika Road lamp glows yellow because of sodium vapour in the lamp, which links to sodium's properties as an element; do not claim more than the lessons taught. Praise new questions as part of science.	Driving Question Board revisit and categorisation. Moving questions from open to answered shows learners the progress of the whole class, makes the link between the phenomenon and each lesson explicit, and validates unanswered questions as the start of future inquiry.	Check that each answer note cites evidence and a lesson, correctly sits under its category, and that Safety answers mention goggles, small quantities, no bare hands, or dose. Count the share of original questions answered (target: at least 80%). Note which questions stay partly answered and carry them forward as follow-up topics.
Model Building Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners compare their Lesson 1 initial model (a drawing or list of why objects are useful) with the completed class family model, a large wall periodic table annotated each lesson. In groups they produce a final model poster: a periodic table sketch showing the alkali metals (Na 2,8,1 with a 1 outer electron label), alkaline earth metals (Mg 2,8,2), halogens (Cl 2,8,7), noble gases (Ne 2,8; Ar 2,8,8) and the transition block (Fe, Cu, Zn), with Al and Pb noted, and arrows from each element to its Kenyan use (sufuria, wire, mabati, salt, Mombasa water, Thika Road lamp, neon signs). Each poster includes a Prediction Rule: 'The position in the periodic table tells us the number of outer electrons, which predicts reactivity and use.'	Say: 'Place your Lesson 1 model next to your final model. What changed?' WAIT TIME 10-15 seconds. Cold-call 3 groups to point to one added idea and the evidence behind it. Ask: 'Using only the periodic table, predict whether potassium would be a good sufuria, and why not.' (K is in Group 1, reactive, more than Na; reactivity $K > Na > Li$.) 'What would you predict for a Group 17 element above chlorine in a similar role?' only if time allows, otherwise skip. Check posters show correct outer electrons and that Na, Cl and NaCl are not confused. Close with: 'Return to Mama Wanjiru: how would you explain to her why her pot and her tap water are safe?' Select 2 learners to answer in under 30 seconds each.	Model comparison (Lesson 1 vs final) with a Prediction Rule. Putting the first and final models side by side makes conceptual change visible, and writing a rule that links position, outer electrons, reactivity and use gives learners a transferable tool for predicting unfamiliar elements.	Look for models that (1) place all five families correctly, (2) link outer electron number to reactivity, (3) connect every anchor object to an element and property, (4) mention safety, cost or availability. Use a 4-point checklist per group. Exit ticket: 'Write one new element job match and defend it in two sentences.' Learners who cannot link position to behaviour get a quick one-to-one with the family table.

	Then learners complete a Then and Now comparison: what did my first model leave out, and what did I add? They end with a short safety and care reflection: one rule that protects self and others and how they would help a classmate with an injury using the first aid kit.			
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners explain sodium's danger and salt's safety using outer electrons rather than just saying 'it changes'? Could they separate free chlorine gas from a 0.5 mg/L dose? How many used the periodic table position as a reason, not just a list of uses? Which DQB questions stayed unanswered and what follow-up could address them (for example, a visit from a Mombasa water works technician or a jua kali artisan from Gikomba)? Were quieter learners and girls given equal turns through cold-calling? Did safety and care for others come through in learners' reflections? Record changes for next time, such as more time for the product pitch.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed and reviewed evidence: sodium floats, fizzes and turns the solution pink in water (alkali NaOH forms); Al is light, Cu conducts very well, Zn coats iron, neon is unreactive; chlorine is a greenish-yellow gas used at about 0.5 mg/L in Mombasa water; sodium vapour gives the yellow Thika Road lamp light.
What did I learn?	We learned that an element's properties depend on its outer electrons and so on its position in the periodic table: Group 1 has 1 outer electron and is very reactive, Group 2 has 2 and is less reactive, Group 17 has 7 and is reactive, Group 18 is full and unreactive. We learned that transition metals are strong, dense conductors, and that safety, cost and availability also affect which element is chosen.
How does this explain the phenomenon?	We explained that sodium is violent alone because its single outer electron is lost very easily, but as NaCl it is stable and safe to eat; chlorine kills microbes at a tiny safe dose; aluminium suits the sufuria, copper suits wire, zinc protects iron on mabati and neon suits signs. The periodic table lets us predict family behaviour and match element to job while balancing safety, cost and availability.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.